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Modeling and Simulation Supporting Material Control and Accounting for Advanced Reactors

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*Keenan Hoffman, Azaree Lintereur, Alexis Maldonado, Philip Lafreniere, Holly Trelue
Los Alamos National Laboratory*

*Jason Tokey, Aditya Savara, Trey Hathaway, Raj Iyengar
Nuclear Regulatory Commission*

**Division of Engineering
Office of Nuclear Regulatory Research
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001**

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Author(s): Hoffman, Keenan Jeffrey; Lintereur, Azaree Tresong; Maldonado, Alexis; Lafreniere, Philip Leo; Trellue, Holly Renee; Tokey, Jason; Savara, Aditya; Hathaway, Trey; Iyengar, Raj

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Modeling and Simulation Supporting Material Control and Accounting for Advanced Reactors

Task 2: Safeguards Modeling and Simulation
Assessment

Los Alamos National Laboratory

Keenan Hoffman, Azaree Lintereur, Alexis Maldonado,
Philip Lafreniere, Holly Trelue

Nuclear Regulatory Commission

Jason Tokey, Aditya Savara, Trey Hathaway,
Raj Iyengar

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Executive Summary

Commercial interest in advanced reactors for power production in the United States is increasing. A variety of advanced reactor designs are being developed, and many of them use non-traditional fuel forms. Therefore, the material control and accounting (MC&A) methods that will be required for these advanced reactor systems also need to be developed to ensure that the necessary safeguards are implemented. To provide information that can be leveraged to explore different MC&A approaches, three representative advanced reactor types were evaluated from a technical safeguards perspective. These reactor designs were selected to encompass common materials and configurations that could affect safeguards considerations. The three designs evaluated were a molten salt reactor (MSR), a gas-cooled reactor (GCR), and a heat pipe-cooled microreactor.

To generate the data needed for this study, high-fidelity 3D reactor system models are created with the Monte Carlo N-Particle® (MCNP®) code to provide nuclear material quantities and other relevant safeguard signatures, including certain simulation uncertainties. Depleted fuel materials can be estimated at various points in time by coupling MCNP with the transmutation code CINDER. With these fission products, SOURCES-4C and Python can generate neutron and gamma source terms, which can in turn be input into MCNP to simulate the radiation signatures that would be present at various points in time.

This simulation framework can be applied to various advanced reactor systems and can serve as an initial estimate for significant nuclear material quantities, signatures, and properties before a system is built. Further, these models can be used to obtain radiation signatures (i.e., gammas and neutrons) from different production mechanisms. This information can be leveraged when assessing material MC&A methods to determine if measured data can be correlated to a parameter of interest. It is also important to consider simulation uncertainty when extracting information from models. Thus, the uncertainty due to different simulation components (e.g. statistics, nuclear data) is tracked and quantified for the different reactor models. Uncertainties associated with different measurement techniques will also need to be considered in MC&A applications.

This work presents an initial modeling and simulation framework to assess MC&A approaches for advanced reactor systems which may not be as conducive to item counting or monitoring. The following example cases explore several safeguards challenges that are not inherent to traditional light-water-reactor systems. The methodology established in this project can be applied to different reactor types or leveraged to provide a more detailed analysis for the development of an MC&A approach specific to one of these models.

Although there are numerous advanced reactor designs being considered, similarities exist in the process required to develop an MC&A approach. For the three advanced reactor designs presented in this report, further model development and simulation efforts should consider the potential signatures that can be correlated to reactor operating parameters. A sensitivity study could also be performed to establish the material changes that are statistically identifiable based on different SNM signatures.

This research report is intended as an exercise paper exploring the use of modelling and simulation for MC&A applications. This report does not make any conclusions or recommendations about the conformance of any specific approach or design with MC&A regulations.

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Acronyms and Abbreviations

Acronym	Definition
AGR	advanced gas reactor
at%	atomic percent
BeO	beryllium oxide
BI	beginning (or book) inventory
BOL	beginning of lifetime
CA	control area
EI	ending inventory
EOL	end of life
GCR	gas-cooled reactor
GLLS	generalized linear least squares
GWd	gigawatt-days
HALEU	high-assay low-enriched uranium
HX	heat exchanger
ID	inventory difference
IPyC	inner pyrocarbon
LANL	Los Alamos National Laboratory
LDRD	Laboratory Directed Research and Development
LiH	lithium hydride
LWR	light-water reactor
M&S	modeling and simulation
MBA	material balance area
MC&A	material control and accounting
MCNP	Monte Carlo N-Particle
MOOSE	Multiphysics Object Oriented Simulation Environment

Acronyms and Abbreviations

Acronym	Definition
MSR	molten salt reactor
MTU	metric ton of uranium
NCERC	National Criticality Experiments Research Center
NRC	Nuclear Regulatory Commission
OPyC	outer pyrocarbon
PWR	pressurized water reactor
SDEF	generalized user-defined Source DEFinition
SEID	standard error on the inventory difference
SiC	silicon carbide
SF	spontaneous fission
SME	subject matter expert
SNM	special nuclear material
TRISO	tri-structural isotopic
UCO	uranium oxycarbide
V&V	verification and validation
VHTR	very-high temperature reactor
wt%	weight percent
YH	yttrium hydride
ZAID	Z and A Identification of isotope, Z-number (number of protons) and A-number (number of nucleons)

1 Introduction

Advanced nuclear power reactors are being designed to address future energy needs and to support non-traditional deployment scenarios. Numerous designs are being considered, ranging from full-scale facilities to compact units that can be transported as fully assembled systems. In addition to various physical designs, a variety of fuel forms and coolants are also being explored. Some of these fuel options include liquid fuel (from which fission products can be removed while the reactor is operating) and tri-structural isotopic (TRISO) particles (which include small, coated fuel kernels to contain fission products as the fuel is irradiated).

Verifying that all nuclear material, regardless of form, is controlled and accounted for is a safeguards requirement that applies to all facilities licensed by the Nuclear Regulatory Commission (NRC). One of the important ways to safeguard nuclear materials is to measure and track the material throughout a facility or fuel cycle. The methods that are used to track and verify special nuclear material (SNM) are part of a material control and accounting (MC&A) program. MC&A programs are well established for commercial light-water reactors (LWRs), but not all of the methods used for LWRs may be applicable for advanced reactors. Determining the techniques that will meet the MC&A requirements for advanced reactors and novel fuel forms remains a challenge; new approaches for MC&A may be required.

The existing MC&A requirements and associated definitions need to be assessed to establish their applicability to some new reactor designs. To adequately analyze different MC&A methods, parameters such as material compositions and associated signatures should be better understood. As these advanced reactors have not yet been physically constructed, modeling and simulations are essential for generating the data needed to identify MC&A challenges and potential solutions.

Monte Carlo simulations can be leveraged to generate material compositions for different reactor designs and operating conditions. To demonstrate this approach, three high-fidelity reactor models of representative, but not proprietary, reactor concepts were modeled using Monte Carlo N-Particle[®] (MCNP^{®*}) code:

1. a molten chloride salt low-enriched uranium-fueled fast spectrum reactor
2. a graphite-moderated gas-cooled reactor
3. a high-assay low-enriched uranium, hydride-moderated, heat-pipe microreactor

These models, which are the foundation for this study, were chosen to illustrate that simulations can be leveraged to generate data for different types of reactors. Such data can include, but is not limited to, quantities of significant nuclear materials, radiation signatures from fresh and activated materials, and reactor-wide parameters (such as reactivity) at different points in time.

A key component of any simulated result is the associated uncertainty. For this application, understanding the source or sources of uncertainty is an important consideration for addressing the potential impact on MC&A metrics (e.g., material inventory). Therefore, this report describes

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Introduction

the simulation approach used, the different codes used to generate data, the models themselves, the sources of simulation uncertainty, and some of the potential material signatures that can be produced by modeling and simulation (M&S). The framework developed as part of this project can be leveraged for future MC&A analyses.

This paper is comprised of five main topics: a description of the methodology; an overview of MC&A considerations; and detailed information about each of the three reactor types.

2 Depletion Codes and Methodology

The first step of this analysis was to generate representative models of the three selected reactor types. These models are used to perform simulations to generate predictions of SNM quantities and associated radiation signatures. Depletion simulations are necessary to determine reactor-fuel composition over time. Gamma radiation predictions are then generated from two different sources: directly from neutron interactions, such as fission, and from the decay of radioactive nuclides. High-resolution depletion simulations are necessary to generate the most accurate gamma spectra from the reactors. Monte Carlo transport codes are well suited for high-resolution calculations as they make fewer assumptions than deterministic counterparts.

Simulations were performed using the Monte Carlo N-Particle transport code MCNP,[®] authored and developed at Los Alamos National Laboratory (LANL). The linkage code, Monteburns, couples MCNP and the isotope generation and depletion tool CINDER'90, and is also authored and developed at LANL[1]. In addition to the MCNP and CINDER'90 codes, SOURCES-4C is also used to enable simulation of neutrons produced from both spontaneous fission and (α , n) reactions. A brief description of the codes used to develop the models and generate the necessary data for an MC&A analysis is provided here.

2.1 The MCNP Code

The MCNP code is a general-purpose radiation transport code developed at LANL over the past forty-six-plus years [2]. MCNP is capable of modeling the fundamental physics of particles as they move through and interact with materials. MCNP has been the international gold standard for particle transport applications for over three decades.

MCNP is used in a variety of fields, including, but not limited to, nuclear criticality safety, reactor physics, medical physics, safeguards and non-proliferation, detector physics, and accelerator physics. World-class verification and validation (V&V) have been performed on MCNP for criticality safety applications. Both static critical and subcritical systems are validated at the National Criticality Experiments Research Center (NCERC). The history of MCNP6 is given in Figure 2-1. It is used in many LANL, and global, radiation transport applications, as shown in Figure 2-2.

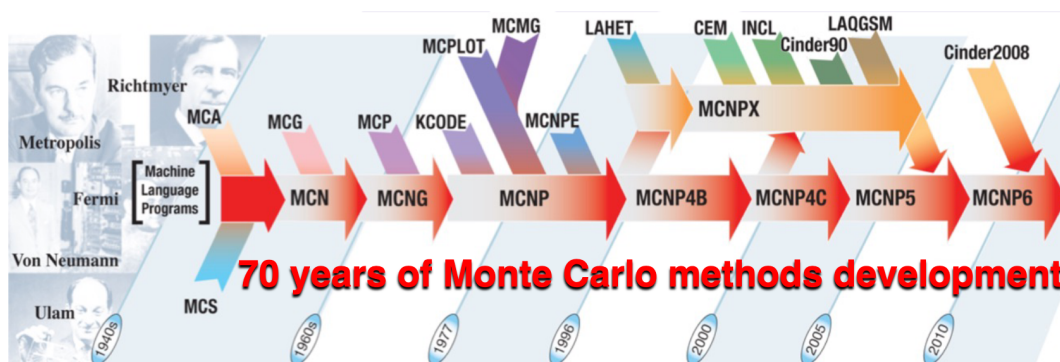


Figure 2-1: MCNP version history

Depletion Codes and Methodology

- Nuclear reactor physics	- Radiation protection and shielding
- Nuclear critical and subcritical experiments	- Radiation detection and analysis
- Nuclear safeguards and nonproliferation	- Nuclear diagnostics
- Criticality safety / manufacturing	- Intrinsic radiation
- Medical and health physics	- Nuclear weapon effects and outputs
- Radiography	- Emergency response / nuclear threat assessments

Figure 2-2: MCNP applications

For historical reasons, MCNP input files are referred to as decks. Lines in the deck are generally referred to as cards, although technically a card can span multiple lines. Materials are defined at the atomic level, using ZAIDs and an atom or mass fraction. The ZAID is a unique identifier used to select data tables. It generally consists of atomic number Z, mass number A, and library specifier ID. For example, 17035.80c is: element 17, isotope 35, and data table 80c. In other words, $^{17}_{35}\text{Cl}$ at room temperature in the ENDF/B-VII.1 Evaluated Nuclear Data Library.

2.1.1 Depletion Calculations

Depletion is a complex process to simulate because the material inventory influences the neutron flux and vice versa. MCNP is internally coupled to CINDER'90 for depletion, and parameters are controlled using the `burn` card. Time is discretized into steps. MCNP sends the neutron flux and reaction rates to CINDER each time step. CINDER then uses these inputs to solve a system of Bateman equations for over 3000 nuclides.

The Bateman equations are stiff differential equations, meaning the different isotope solutions vary at dramatically different speeds. This would normally require restrictively small time steps to tailor to the quickest production and loss terms. However, most depletion codes use some sort of predictor-corrector scheme to allow for longer steps to be taken. One step of the predictor-corrector routine used by MCNP is outlined below:

1. An initial MCNP criticality calculation determines depletion parameters: k_{eff} , 63-group flux, energy-integrated reaction rates, fission multiplicity (ν), and recoverable energy per fission (Q value) [2]. This information is sent to CINDER'90.
2. CINDER'90 performs a “predictor” depletion step, using the information provided by MCNP, to deplete the specified materials for one-half of the provided step duration.

Depletion Codes and Methodology

3. A time-independent MCNP calculation is performed using the material composition provided by CINDER'90. This produces depletion parameters for $t_{n+\frac{1}{2}}$ (average reaction rates), which are again sent to CINDER'90.
4. CINDER'90 then performs a “corrector” depletion step. Material depletion is calculated for the full time duration using average MCNP reaction rates from the halfway point.
5. Another time-independent MCNP calculation is done to determine k_{eff} after the material depletion and to calculate the initial flux for the next time step.

Although the built-in **burn** card is relatively easy to use, it is up to the user to select the correct temporal and spatial resolution to achieve the desired answer. Across a single time step, material concentrations and flux are treated as constant. The predictor-corrector algorithm eases this constraint slightly, allowing for linear changes across the step at the cost of two MCNP calculations for each time step.

MCNP flux tallies are normalized “per source particle,” representing the fraction of the total flux in each region. In order to calculate the unnormalized flux (needed for depletion), the normalized flux is multiplied by the number of source particles, which is heavily influenced by reactor power:

$$\phi_{total} = \phi_{MCNP} \frac{P\nu}{(1.602 * 10^{-13} \text{ J/MeV}) Q} \quad (1)$$

In Equation 1, P is the power (in watts), ν is the average number of neutrons produced per fission, and Q is the recoverable energy (in MeV) per fission event [3].

Another parameter influencing simulation resolution is the number of tracked isotopes included in the MCNP transport calculation. Not all 3000+ isotopes produced by CINDER have the complete set of transport cross-sections required by MCNP. There are less than 300 isotopes that can be tracked, and these isotopes influence the flux that is sent to CINDER to transmute the inventory of 3000. For tracked isotopes, a one-group reaction rate is sent directly. For the rest, a 63-group flux is used with the internal CINDER transmutation library and group collapse is performed. In general, the 3000+ nuclides are not essential to the MCNP simulation, as these nuclides are short-lived and do not impact the neutronics significantly.

Unmodified, MCNP depletion results only include tracked isotopes. In order to generate a gamma spectrum using the full inventory produced by CINDER, a different tool is needed to generate material inventory. Historically, Monteburns [1] would be used for this purpose, as it allows more control over the depletion process. However, certain drawbacks to this workflow are discussed below and motivated a second workflow, which was tested on two of the reactor concepts.

2.1.2 Monteburns

Monteburns is an external depletion coupling code which links MCNP to the isotope production and depletion code CINDER'90 (Figure 2-3). CINDER'90 tracks the evolution of thousands of isotopes over time, including fission products and actinides, which allows further generation of system parameters, such as neutron and gamma emissions, for nondestructive assay safeguards assessments [4][5].

Depletion Codes and Methodology

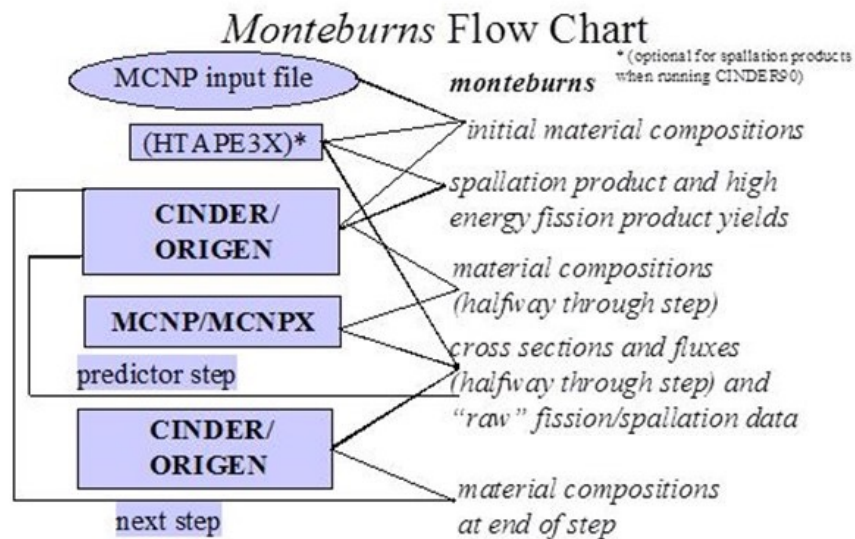


Figure 2-3: Monteburns flow chart

Monteburns provides additional control over the depletion calculation that is not available using the internal `burn` card. Primarily, the complete CINDER nuclide inventory can be printed to the output file. Other examples include the ability to select how many CINDER'90 "internal" burn steps are performed between each MCNP calculation.

Once the material composition is obtained, gamma spectra from fission products can be created as source term inputs for the MCNP input file using a separate post-processing script [5]. Monteburns was used to perform the depletion calculations for the MSR as it was the only tool available that could access the 3000+ nuclides from CINDER at the time the work began. Once this feature was added to a developer version of MCNP, the other two reactor systems were analyzed using the new tool.

2.1.3 Custom MCNP Depletion

MCNP can be ported to new machines more rapidly than Monteburns and does not require external coupling to another code. Fewer scripts and input files are required, making it easier to automate with codes like Python. In order to output the 3000+ isotopes produced by CINDER'90 using the built-in depletion solver, a custom MCNP `burn` card was developed by the X-Computational Physics division (XCP-3) at LANL using Laboratory Directed Research & Development (LDRD) funding. The workflow shown in Figure 2-4 was used to generate gamma sources from decaying fuel for the gas-cooled reactor (GCR) and microreactor.

Depletion Codes and Methodology

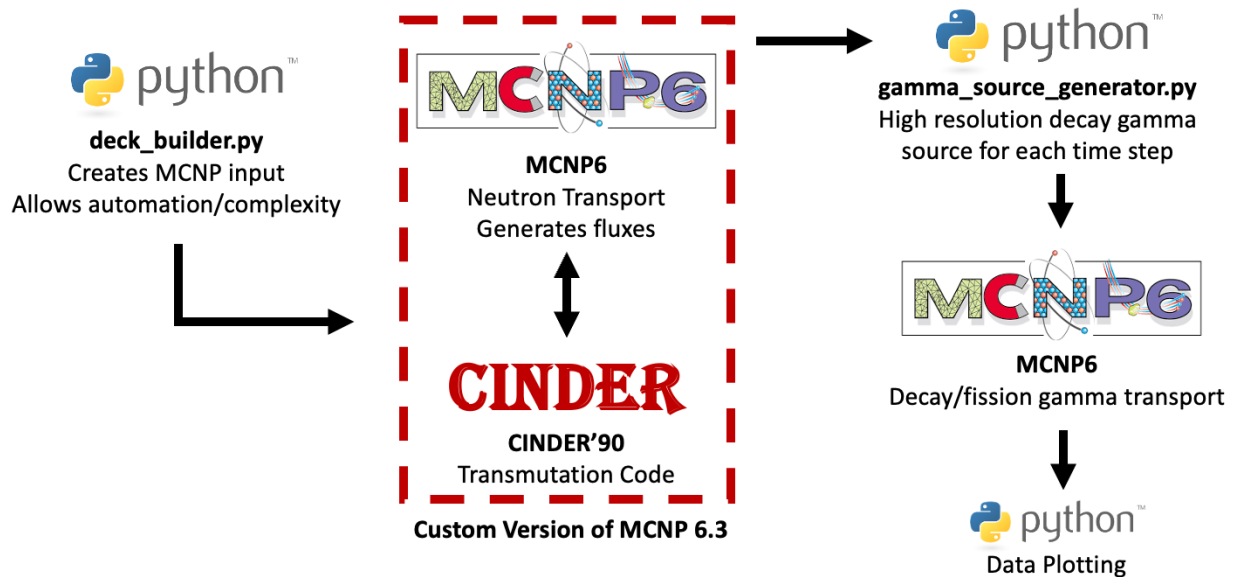


Figure 2-4: Gamma source generation and transport workflow

Although Python is not required to create the MCNP input, using Python allows great flexibility in creating iterative studies quickly. The depletion calculation (red box) is then performed to determine material composition over time. Material compositions are then input into a modified source generation script to calculate the gamma energies and intensities produced by the decaying fuel. Generated gammas can then be transported in MCNP to simulate material interactions and detector responses.

The process greatly streamlines decay gamma analysis and allows more complex models to be created quickly with less manual intervention. For example, fuel can be removed from the simulation at any time step, and the gamma detector response can be determined over time. This type of analysis can be useful for safeguards analyses for advanced reactor systems. Additionally, prominent gamma decay lines can be traced back to individual nuclides, potentially yielding relevant safeguard information.

2.2 SOURCES-4C

In addition to induced fission neutrons from the reactor, decay neutrons are also associated with spent reactor fuel. One such neutron source is the (α, n) reaction, in which an alpha particle is captured by the nucleus and subsequently releases a neutron. Another neutron source is the spontaneous fission reaction, which differs from induced fission in that it does not require an incident neutron to undergo the reaction.

Decay neutrons are calculated using the code SOURCES-4C. This is a code written in FORTRAN that calculates (α, n) and spontaneous fission neutron source magnitudes and distributions [6]. The input to SOURCES-4C is the compositions of actinides and low (α, n) targets, and then

Depletion Codes and Methodology

the neutron source rates and distributions, in units of n/s-unit volume, are output. Calculations are made using data libraries for (α, n) cross-sections, (α, n) branching libraries, and spontaneous-fission-decay source data. A Python script is applied to the output source rates and energy distributions to produce an SDEF (generalized user-defined Source DEFINition) card that can be integrated with the MCNP model. For this work, only the molten salt reactor (MSR) conceptual system was analyzed with SOURCE-4C to provide a proof-of-concept for the methodology. This workflow could be applied to the other reactor designs.

The uncertainty associated with the simulated predictions impacts their value for assessing MC&A approaches. Thus, different sources of uncertainty are discussed in Section 3.

3 Sources of Uncertainty

Uncertainty quantification is an important consideration when assessing an MC&A approach. If data is generated from simulations, then the uncertainty associated with those simulations needs to be understood.

Thus, several sources of uncertainty are assessed in this work: statistical uncertainty (from random sampling), nuclear data (input parameter uncertainty), depletion durations (model approximations which affect the simulation's material composition and depletion amounts during the simulation). Parameter and model approximation uncertainties may be able to be reduced in the future with improved nuclear data uncertainties and higher-fidelity models. Additional information for each of these categories is provided here, and the respective contributions are discussed for each of the reactor models in Sections 5–7.

3.1 Monte Carlo Standard Deviation, Confidence Intervals

MCNP is used to perform criticality simulations. MCNP executes many simulations to gain a statistical sampling of particle "histories," where each history is a simulation of the life-cycle of a particle. The particle histories interact with the system material via random sampling from probability distributions via radiation interactions (such as collision, capture, fission, etc). By simulating numerous particle histories, an expected outcome can be determined for a system. In MCNP, quantities of interest (e.g. neutron and gamma flux) are recorded from the simulations using "tallies," which count the number of expected particle outcomes (such as crossing a geometric surface). This enables estimates for system parameters, such as flux at a reactor power level and radiation energy depositions for each reactor material.

As described in Section 2.1, MCNP was used in this work to generate criticality simulations. Multiple simulation cycles are performed to ensure k_{eff} has converged to a solution. Using the statistical distributions from the cycles, and given the central limit theorem, MCNP calculates the mean and confidence intervals to represent expected k_{eff} and a statistical uncertainty related to finite sampling (as described in Section 2.6.3 of MCNP Code Version 6.3.0 Theory & User Manual, Document ID LA-UR-22-30006). These 1σ confidence intervals are reported in this report as "Statistical Uncertainty" for k_{eff} .

Once the k_{eff} has converged to a solution after multiple cycles, the MCNP calculation is considered a snapshot of the neutronic behavior of the core at that point in time. These expected values, their associated confidence intervals, and their relationship to random sampling is well understood. These confidence intervals are proportional to the inverse of the square root of the number of samples [2], and thus approach zero for very large sample sizes.

For typical nuclear reactor designs with large number of samples (greater than 10 million), standard deviations in k_{eff} are typically lower than 0.05 percent or 50 percent mille (pcm) of reactivity. From MCNP calculations, confidence levels can be calculated for a variety of quantities of interest, including various flux tallies and reaction rates.

Sources of Uncertainty

3.2 Nuclear Data Uncertainty

Nuclear data uncertainty is a type of epistemic uncertainty. The inclusion of nuclear data uncertainties on the effective multiplication factor is a more recent development of uncertainty quantification in nuclear engineering. Current nuclear data uncertainties commonly available to the public are detailed in the Evaluated Nuclear Data Library VII.1 (ENDF/B-VII.1) [7]. These nuclear data uncertainties are routinely evaluated, updated, and discussed in the nuclear engineering community. The uncertainties, like the nuclear data itself, come from a combination of experimental and theoretical data. Nuclear data uncertainties can also be correlated, for example, uncertainty in one nuclear data parameter may be magnified because its calculation relies on a different nuclear data parameter. In this work, the nuclear data uncertainties are considered separately and independently from the uncertainties described in Section 3.1.

Nuclear data uncertainty impacts systems differently. For example, two nuclear systems with different materials will have varying nuclear data uncertainties. Furthermore, even two systems with similar nuclear materials can have differences in nuclear data-induced uncertainties if the materials are arranged differently, causing changes in neutron spectra. As a consequence, sensitivity coefficients with respect to incident neutron energy must be generated individually for every nuclide in each system with a neutron transport simulation.

A single Monte Carlo simulation can produce all the necessary inputs to generate nuclear data-induced uncertainty for a system using a statistical analysis code. For example, Whisper [8] employs a generalized linear least squares (GLLS) Bayesian analysis methodology to reduce data-induced uncertainties using nuclear criticality benchmarks with relevant materials and energy spectra [9]. Nuclear data-induced uncertainties for the effective multiplication factor (k_{eff}) typically range between 100 to 500 pcm [10].

The existing nuclear data uncertainty quantification framework with Whisper is for the effective multiplication factor. Although impacts to radiation flux tallies can be inferred, additional research and development is needed to directly quantify nuclear data uncertainties for flux tallies.

3.3 Material Composition Uncertainty

Uncertainty in material composition can occur from inadequate discretization of the depleted materials or inadequate number of time steps. Discretization error is assessed through convergence studies and can occur from insufficient time steps, spatial region approximations [11], or imperfect fission-product tracking. Uncertainty in the nuclear data also affects the reaction and transmutation rates at each depletion time step. Propagating this uncertainty through a Monte Carlo simulation is an area of ongoing research [12]. Uncertainty due to material composition can vary significantly and can be assumed to be at least 500 pcm for k_{eff} .

Quantifying this uncertainty for a system requires input from subject matter experts (SMEs), who can identify common sources of uncertainty on a case-by-case basis. For example, the MSR depletion simulation requires a different depletion approach than the microreactor; because the MSR has flowing molten salt with fuel, the molten salt will need to “move” in the depletion simulation, as is done in this work. The microreactor, however, has static fuel, i.e., movement of the fuel is not necessary because most microreactor concepts will not include traditional refueling intervals.

Sources of Uncertainty

The neutronic solution can also be influenced by other physical processes such as thermomechanics, thermal hydraulics, and fluid dynamics [13]. Outside of the simulation space, uncertainties in detectors and any tool used to measure a signature should also be considered when determining the feasibility of using a radiation transport simulation for nuclear safeguards.

3.4 Depletion Validation

Nuclear fuel depletion is a complex process in which changes in neutron flux and material composition have a significant impact on each other. In this work, simulations approximate time-dependent behavior by performing a series of steady-state calculations in steps. To compensate for the extreme variation in isotope half-lives, multiple steady-state predictor-corrector simulations are typically used for a single time step to achieve more accurate results [14]. Even so, significant errors in isotope mass are difficult to avoid because of the large number of pathways, and uncertainty can influence the calculation.

Fortunately, plutonium isotopes have simple production pathways with relatively well-known neutron cross-sections. For example, MCNP6 accurately predicted plutonium isotopes in the Robinson benchmark within 10 percent accuracy in most cases [15]. Most Monte Carlo coupled depletion codes are capable of predicting important actinides with 15 percent accuracy or better [16].

In Sections 5–7, each reactor system will be discussed in detail, including a preliminary uncertainty quantification analysis.

4 Material Control and Accounting

Models can be used to obtain information about the expected performance of advanced reactors and predict changes in SNM inventory. This information can be leveraged for a variety of applications, including the development of MC&A approaches. Facilities licensed by the NRC must abide by certain MC&A regulations. These regulations are intended to ensure that SNM information is accurate, authentic, and sufficiently detailed to enable a licensee to maintain current knowledge of its SNM and manage its program for securing and protecting SNM from any loss, theft, diversion, or misuse [17].

The requirements for MC&A, together with those for physical protection of facilities and information security, make up the primary elements of the NRC's SNM safeguards program. The MC&A component of the larger SNM safeguards program helps ensure that SNM is not stolen or otherwise diverted from the facility and supports the NRC's strategic goal of ensuring the secure use of radioactive materials.

MC&A methods and requirements for traditional nuclear facilities, including LWRs, are well established. However, the implementation of MC&A programs for advanced reactors could require different techniques and approaches. The NRC regulations for MC&A of SNM, 10 CFR Part 74, will be used as the primary basis of this discussion [18].

4.1 Material Categories

Nuclear material categories are used to define different levels of risk. Depending on the facility type, some of the MC&A requirements are category dependent. Current U.S. LWRs are considered item facilities and are granted exemptions from some of the material category-specific MC&A requirements. However, at the time of writing, it has not been established if the material category-specific MC&A requirements will apply to all (or any) of the advanced reactor designs being developed. For context, the basic material category definitions are provided in Table 4-1.

Material Control and Accounting

Table 4-1: NRC defined material categories

Material	Category I	Category II	Category III
Pu	>2 kg	>500 g	≤500 g and >15 g
²³⁵U	≥5 kg ²³⁵ U (contained in uranium enriched to 20% or greater)	>1 kg ²³⁵ U (contained in uranium enriched to 20% or greater)	≤1 kg and >15 g ²³⁵ U (contained in uranium enriched to 20% or greater)
		<i>or</i>	<i>or</i>
		≥10 kg ²³⁵ U (contained in uranium enriched to ≥10% but <20%)	the combination of ²³⁵ U+Pu+ ²³³ U<10 kg and >1 kg ²³⁵ U (contained in uranium enriched to 10% or greater but <20%)
²³³U	≥2 kg	>500 g	≥10kg ²³⁵ U (contained in uranium enriched above natural but <10%) ≤500 g and >15 g

4.2 Nuclear Material Inventories

NRC safeguards regulations include requirements for performing material inventories. These requirements are based on the type of facility and the material category. For item facilities, the items must be tracked. For bulk facilities, the quantity of material present must be confirmed. For current reactors, a sealed core is considered to be one item for inventory purposes. Some advanced reactor designs may not be conducive to item accounting. As such, these case studies consider the requirements for item and bulk material facilities. The units that are considered items must be defined and it is expected that no items will be unaccounted for. Due to the nature of bulk material, and bulk material measurements, discrepancies between the expected and measured amounts are expected. However, the regulations stipulate permissible limits on the discrepancies. There are also uncertainties associated with different measurement techniques. Thus, there will be uncertainties associated with the bulk material measurements.

As mentioned above, traditional LWRs are item facilities, but the unique features and fuel forms for some of the advanced reactor designs may result in facilities that are bulk, or a combination of item and bulk. This could influence the applicability of some of the current safeguards requirements. Further, for advanced reactors the fuel form, access, available signatures, etc., will all influence the measurement techniques that can be applied. Thus, for some situations, such as measuring radiation signatures on flowing material, a large amount of uncertainty could be associated with the results.

The information that is available, and ultimately leveraged, to determine MC&A metrics (e.g., inventory) needs to be understood to adequately assess different options and approaches. For example, measured radiation signatures are produced by different mechanisms, so if the total neutron count rate from a sample is used as a data source for SNM mass predictions, the results will be inaccurate if the neutron production mechanisms (i.e., spontaneous fission, induced fission, (α, n)) are not taken into consideration. Simulations can be used to estimate these different contributions to

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the total radiation signal and to establish what measurement techniques can be used to extract the necessary information. For validation, any simulation predictions need to be empirically confirmed once a system is built or with a representative experiment.

MC&A-related considerations for each of the reactor models are discussed in the upcoming sections. These discussions provide a preliminary overview of aspects related to modeling and simulation can be used to support MC&A for advanced reactors. Additional research would be required to understand the impacts of simulation uncertainty on the ability to use models as a means of establishing expected signatures under different operating conditions. The use of simulation may aid in improved quantification of measurement uncertainties and assessment of different measurement options.

5 Molten Salt Reactor

The MSR design modeled in this study is a conceptual fast spectrum molten-salt chloride reactor derived from several designs but not matching any established MSR design [19][20][21]. This representative MSR design allows for exploration of safeguards challenges present with a liquid fuel material without having to disclose or assess the details of a proprietary design.

The MSR is simulated to operate at 2,000 MWth. The uranium enrichment is 19.5 weight percent (wt%) ^{235}U . The percentage of uranium in the sodium chloride salt is 17.5 atomic percent (at%) (see composition in Table 5-1).

Table 5-1: Uranium Molten Chloride Fuel Composition

Isotope	Atom Fraction
^{235}U	0.0348
^{238}U	0.1402
^{35}Cl	0.4180
^{37}Cl	0.1320
^{23}Na	0.2750

Throughout the depletion calculation, fuel volumes are shuffled through five regions to emulate the flow of molten salt up through the salt loop. In this simulation, the salt moves from the core exit to the heat exchanger and then to the core entrance. Spent fuel salt removal and unirradiated fuel salt addition occur in the heat exchanger volume.

5.1 Geometry

The assumed geometry of the MSR for this study is given in Figure 5-1. For the MSR simulation performed here, the uranium-fueled core has an outer radius of 200 cm and is 375 cm in height. The fuel is surrounded by a 50 cm stainless steel vessel. The total fuel volume present in both the core and the heat exchanger is about $6.15 \times 10^7 \text{ cm}^3$ (61.5 cubic meters).

An example of how the fuel salt flows through the MSR system and the heat exchanger in the five-region configuration is also shown in Figure 5-1. Each fuel salt MCNP material has a unique ID and flows through each region (A, B, C, D, E, and HX) during depletion every 10 days to approximate realistic fuel movement inside an MSR.

Molten Salt Reactor

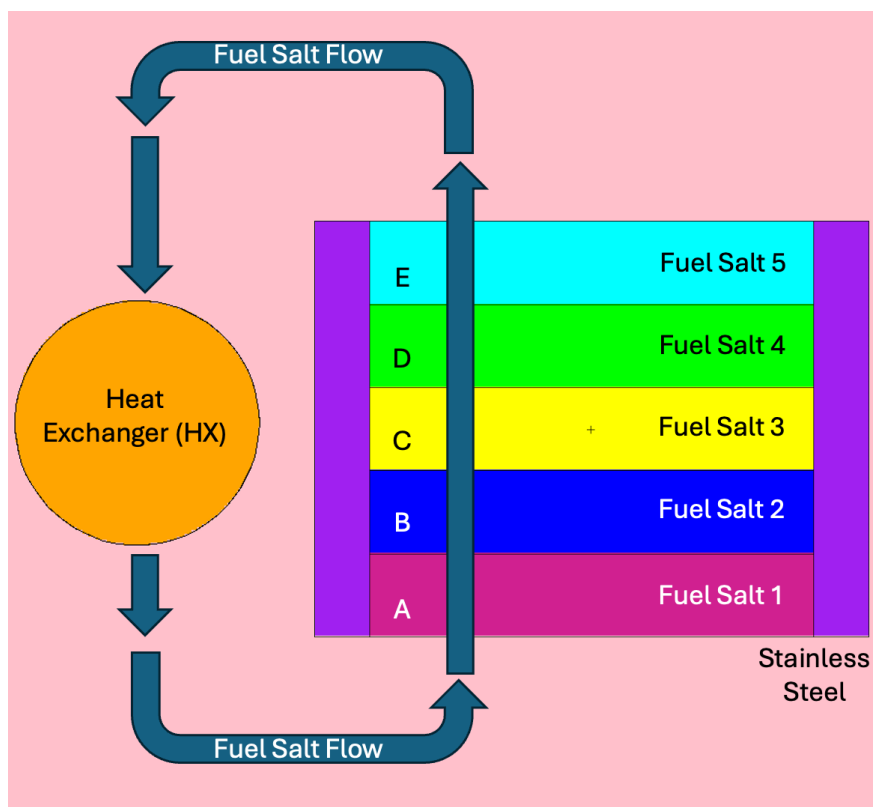


Figure 5-1: MSR MCNP geometry and fuel salt flow using five depletion regions

5.1.1 Model Assumptions

For all MSR simulations in this study that use more than one depletion material, molten salt material movement is simulated as starting at the bottom of the core and slowly moving upwards over time. In an actual molten salt reactor this circulation occurs over a period of several hours, but in MonteBurns materials are shuffled every 10 days. Specifically, the molten salt starts at the bottom of the core and moves slowly to the top and then outward to the heat exchanger and finally back into the bottom of the core (Figure 5-1). Six material movements occur over the span of 60 days, at which point the fuel material returns to the original starting position. The depletion simulation was run for 1000 days, or about 23 gigawatt-days per metric ton of uranium (GWd/MTU).

The geometry approximates the external heat exchanger-region as a large sphere located outside the stainless steel cylinder holding the liquid fuel (any effects of the heat exchanger geometry are negligible in this work). In a true system, pipes would connect the core and heat exchanger regions. Little neutron flux is incurred in this heat exchanger-region, so the exact geometry and location does not influence the core neutron results; the goal of including this sphere is to establish a region where fuel is added and fission products are removed. Removal of most fission products (atomic numbers 21 through 79) and addition of more high-assay low-enriched uranium (HALEU) uranium (^{235}U and ^{238}U) at the rate of 12 kg every 10 days occurs in this external region before the molten

salt is inserted back into the core.

5.2 Burnup

The reactor was modeled with three different levels of fidelity to determine how much spatial depletion resolution is necessary to obtain accurate isotopic results as a function of burnup.

1. The core was modeled as one region with no fuel movement.
2. The core was split into five equal-volume axial segments with a heat exchanger. Fuel was discretely moved between the axial segments and heat exchanger at regular intervals throughout the calculation.
3. The core was segmented into five radial and five axial regions plus heat exchangers and holding regions to capture even more spatial fidelity.

The difference in fuel isotopics resulting from spatial depletion resolution is shown in Table 5-2. Significant differences were observed in higher plutonium isotopes. The differences occur because the neutron energy spectrum varies with position, which affects the ^{239}Pu neutron capture reaction rate and the production of ^{240}Pu and higher isotopes. The relative percent error when going between one and five regions is as large as 15 percent, whereas the relative percent errors going from five to twenty-five regions were only a couple of percent. Thus, five regions were used for this modeling exercise, which provides sufficient accuracy to assess the scale of other uncertainty contributions for the simulations performed.

Table 5-2: Percent difference in actinide mass with 1 vs. 5 depletion regions

Isotope	1 vs. 5 regions	5 vs. 25 regions
^{234}U	+0.64%	+0.90%
^{235}U	−0.21%	+0.20%
^{236}U	+0.42%	+0.38%
^{238}U	+0.00%	+0.28%
^{237}Np	−6.88%	−1.55%
^{239}Pu	+0.72%	+0.20%
^{240}Pu	−15.36%	−3.73%

The five-region case is used to calculate isotopic mass compositions as a function of burnup using 100 10-day time steps. The relative simplicity of the model allowed for much higher temporal fidelity compared to the reactors with explicit TRISO fuel. For the five-region case with circulating fuel and removal of fission products and addition of uranium, results are plotted in Figure 5-2 and given in Table 5-3.

Figure 5-2 shows the change in ^{235}U and ^{239}Pu mass over time. Results were plotted every ten time steps. Uranium mass decreases as it is burned by fission at a higher rate than fuel is added.

Molten Salt Reactor

Plutonium concentration increases in the system over time as ^{238}U is transmuted after a neutron capture.

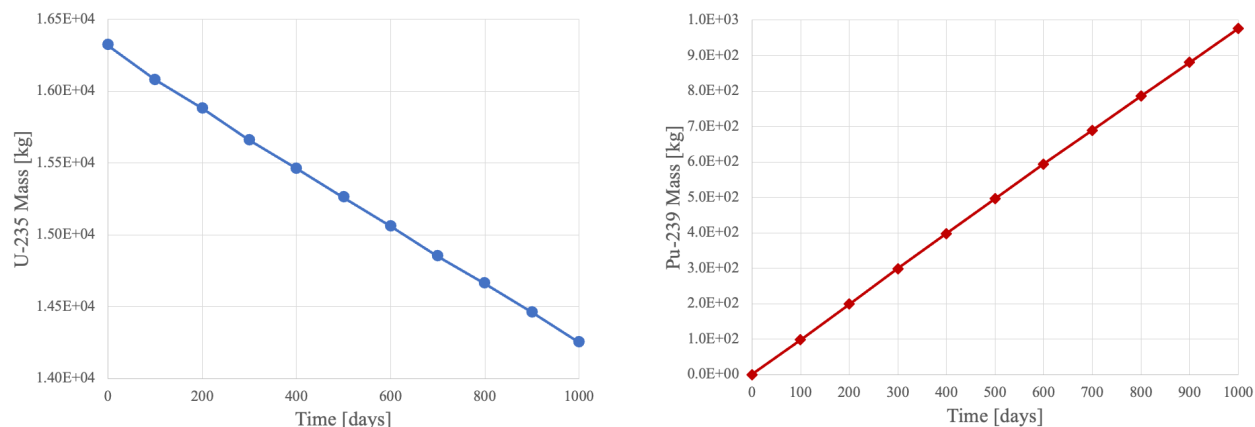


Figure 5-2: U-235 and Pu-239 mass with respect to time

Table 5-3: MSR actinide mass (in grams) as a function of burnup for 5 region case

Time (days)	^{234}U	^{235}U	^{236}U	^{238}U	^{237}Np	^{238}Pu	^{239}Pu	^{240}Pu	^{241}Pu
0	0.000e0	1.632e7	0.000e0	6.660e7	0.000e0	0.000e0	0.000e0	0.000e0	0.000e0
100	1.184e2	1.608e7	4.698e4	6.660e7	6.060e2	2.195e0	9.800e4	1.457e2	1.501e-1
200	2.353e2	1.588e7	9.370e4	6.660e7	1.378e3	1.055e1	1.9880e5	5.219e2	1.029e0
300	3.506e2	1.566e7	1.398e5	6.660e7	2.260e3	2.604e1	2.9870e5	1.128e3	3.268e0
400	4.641e2	1.546e7	1.856e5	6.640e7	3.249e3	4.977e1	3.9790e5	1.973e3	7.560e0
500	5.763e2	1.526e7	2.310e5	6.620e7	4.344e3	8.260e1	4.9620e5	3.052e3	1.450e1
600	6.860e2	1.506e7	2.761e5	6.600e7	5.547e3	1.253e2	5.9370e5	4.370e3	2.474e1
700	7.970e2	1.485e7	3.208e5	6.600e7	6.850e3	1.791e2	6.8900e5	5.917e3	3.899e1
800	9.040e2	1.466e7	3.651e5	6.600e7	8.270e3	2.451e2	7.8600e5	7.740e3	5.789e1
900	1.010e2	1.446e7	4.089e5	6.600e7	9.800e3	3.238e2	8.8100e5	9.770e3	8.190e1
1000	1.116e2	1.425e7	4.524e5	6.600e7	1.143e4	4.168e2	9.7601e5	1.205e4	1.122e2

5.3 Uncertainties

Table 5-4 below shows the Monte Carlo and nuclear data standard deviations (σ) every ten depletion time steps. The variation between Monte Carlo trajectories is not the dominant effect, as shown in Table 5-4. The MCNP (Monte Carlo) sampling uncertainty in k_{eff} is 0.00012 or 12 pcm (average of the eleven time steps presented in Table 5-4), and the average uncertainty resulting from nuclear data in k_{eff} is 0.0051 or 510 pcm.

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Table 5-4: MSR burnup, k_{eff} , uncertainties, and percentage of neutron energies causing fission

Time [days]	Burnup [GWd/MTU]	k_{eff}	Statistical k_{eff} σ	Nuclear Data k_{eff} σ	% thermal fissions	% epitherm. fissions	% fast fissions
0	0.00000	1.033881	0.000165	0.00534	0.00%	41.42%	58.58%
100	2.41725	1.032251	0.000117	0.00502	0.00%	41.51%	58.49%
200	4.83440	1.030758	0.000118	0.00517	0.00%	41.61%	58.39%
300	7.25155	1.028863	0.000113	0.00515	0.00%	41.66%	58.33%
400	9.66860	1.027456	0.000113	0.00506	0.00%	41.74%	58.26%
500	12.08571	1.025492	0.000116	0.00507	0.00%	41.81%	58.19%
600	14.50305	1.024087	0.000113	0.00509	0.00%	41.90%	58.10%
700	16.92029	1.022210	0.000117	0.00489	0.00%	41.95%	58.05%
800	19.33761	1.020384	0.000116	0.00514	0.00%	42.03%	57.97%
900	21.75489	1.018523	0.000120	0.00507	0.00%	42.10%	57.90%
1000	24.17208	1.017571	0.000119	0.00505	0.00%	42.16%	57.84%

Figure 5-3 presents the nuclides that contribute the top sources of nuclear data uncertainty in k_{eff} for all possible neutron interactions. Recall that for the ZAID identifier, the first two numbers are the proton count, and the last three numbers are the nucleon number. For example, 17035 is element 17 (chlorine) isotope 35, i.e., $^{35}_{17}\text{Cl}$. Pu-239 plays a significant role in the nuclear data-induced uncertainty as it is produced during the simulation, as shown in Figure 5-3,.

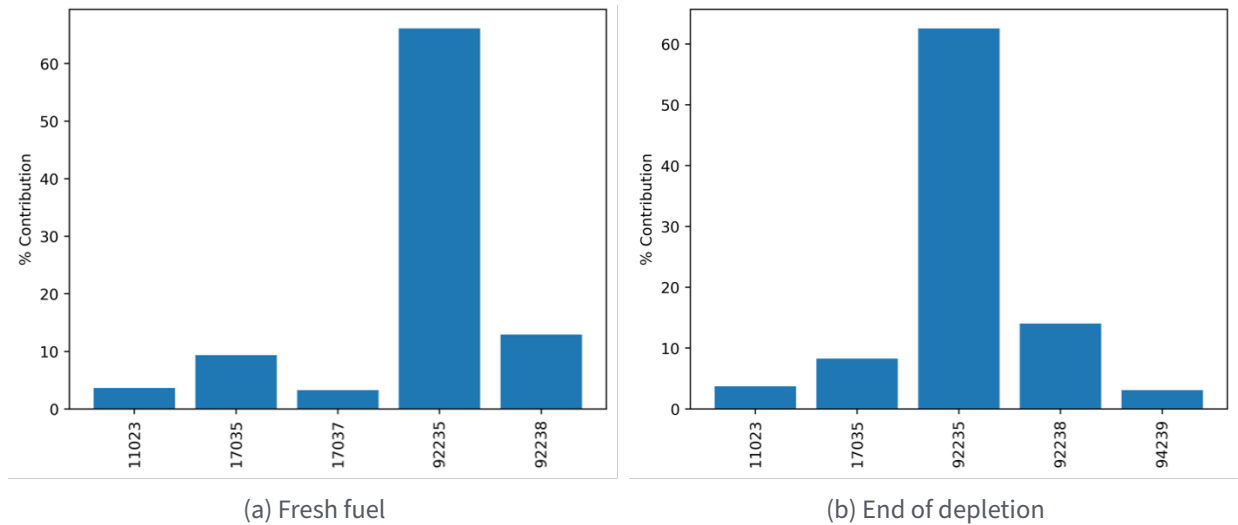


Figure 5-3: MSR top five isotopes contributing to nuclear data induced uncertainty in k_{eff}

Figure 5-4 goes into additional detail, listing both the ZAID and the neutron interaction that contribute to the nuclear data-induced uncertainty to k_{eff} . The ^{235}U radiative capture reaction, in which a neutron is absorbed and the nucleus generates a gamma to stabilize, is the primary

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contributor to the nuclear data-induced uncertainty to k_{eff} , with over 50 percent of the share as shown in Figure 5-4. Although ^{235}U comprises less than 20 percent of the uranium by mass, it is the largest contributor to uncertainty due to its significant role in the fission process.

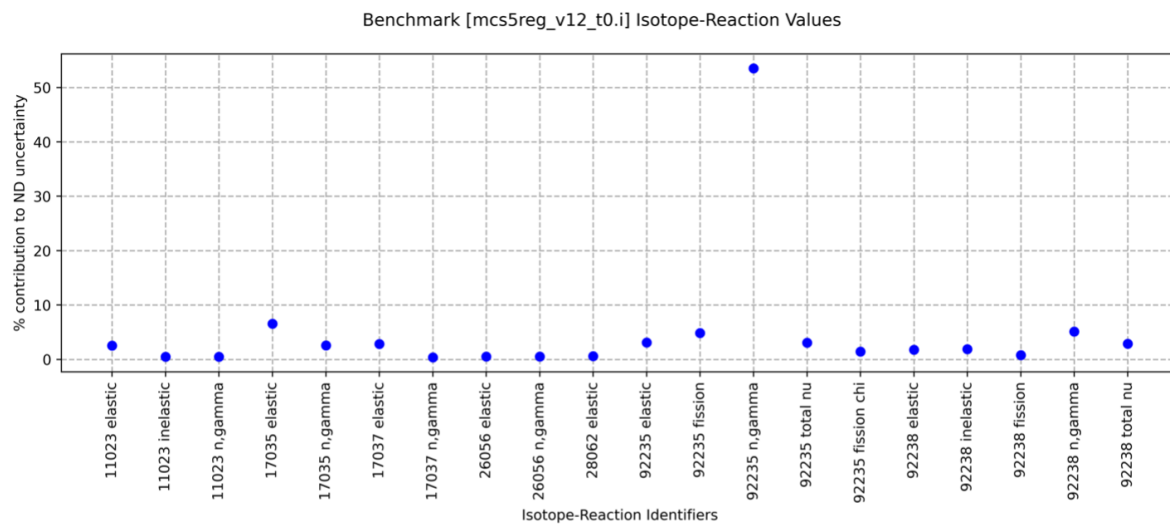


Figure 5-4: MSR top twenty nuclide interaction uncertainty contributors to k_{eff} with fresh fuel

Plotting the cross-section with uncertainty for this nuclide interaction (n, γ), as shown in Figure 5-5, illustrates that this uncertainty most likely stems from the epithermal incident neutron energies, which can have uncertainties ranging over several barns.

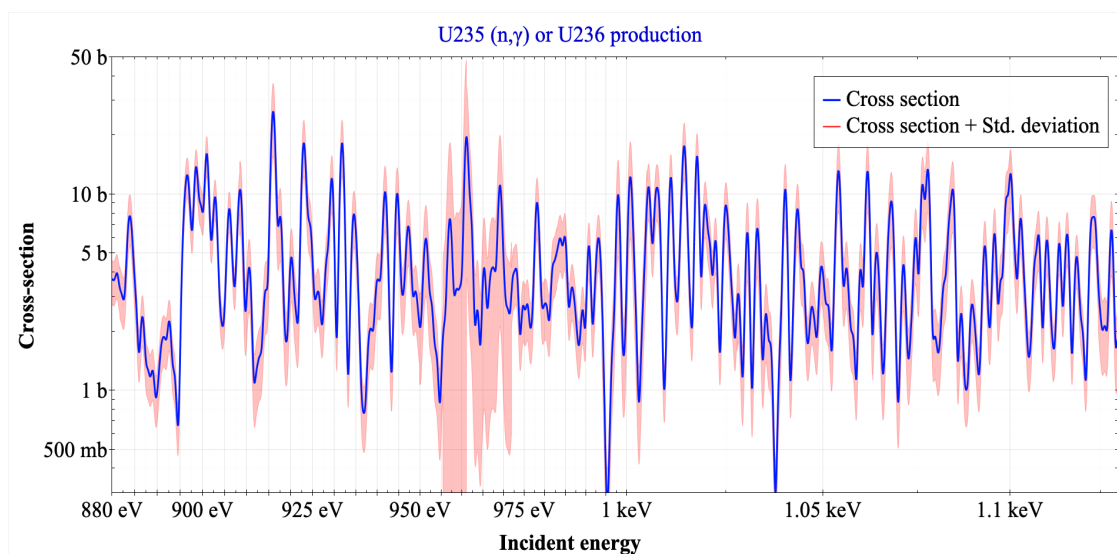


Figure 5-5: Radiative capture (n, γ) U-235 cross-section at epithermal neutron energies, plotted with JANIS [22]. Red solid line is the cross section and green transparent color represents the uncertainty

Figure 5-6 shows the (n, γ) cross-section and uncertainties for the fast region, which only varies a fraction of a barn and is more predictable than the resonances found in the epithermal neutron energy region.

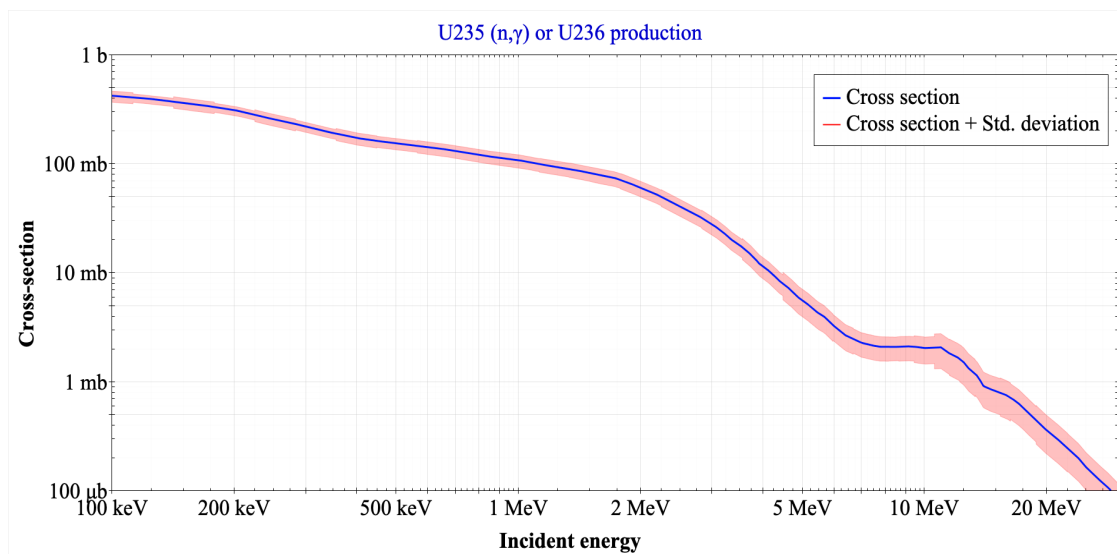


Figure 5-6: Radiative capture (n, γ) U-235 cross-section at fast neutron energies, plotted with JANIS. Red solid line is the cross section and green transparent color represents the uncertainty

The MSR design has the highest nuclear data-induced uncertainty of the concepts studied, most likely due to the large percentage of epithermal neutrons causing fission, as shown in Table 5-4. Because a significant source of uncertainty to k_{eff} is due to the (n, γ) cross-section, this is also likely a large source of uncertainty for gamma flux and dose tallies in the MCNP neutron/gamma coupled simulation. Because the gammas depend on this cross-section, nuclear data uncertainty will impact the gamma flux presented in Figure 5-8 as described in the next section.

5.4 Neutron and Gamma Counts

The neutron and gamma spectra from the MSR were calculated with fresh fuel and at the end of depletion. Neutrons and gammas can be produced from activation, radioactive decay, and fission processes; they are generated both while the reactor is operating and after it is shut down.

Figure 5-7 compares the neutron fission and decay spectra in the core with fresh fuel and with fuel at end of depletion while the reactor is operating. The fluxes have similar shapes because of neutrons interacting with the salt and structure of the MSR with varying levels of magnitude.

Figure 5-8 shows the gamma spectra with fresh fuel and with fuel at the end of depletion while the reactor is operating.

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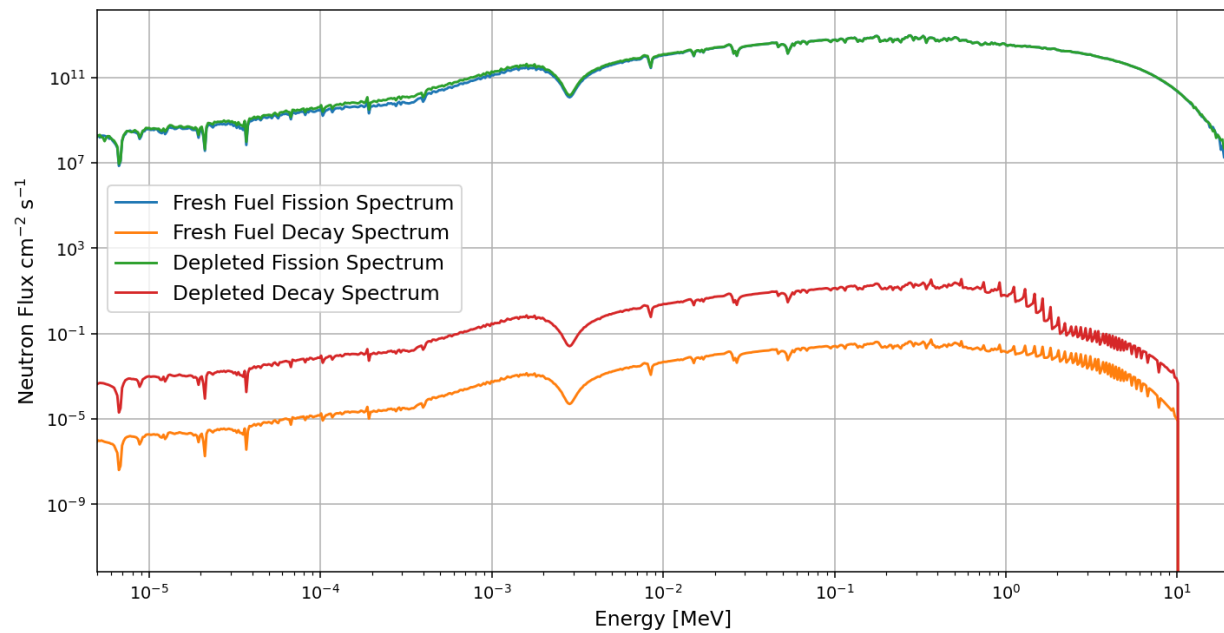


Figure 5-7: MSR neutron fission spectra with different fuel compositions. The two fission spectra are of the same magnitude and appear overlapped.

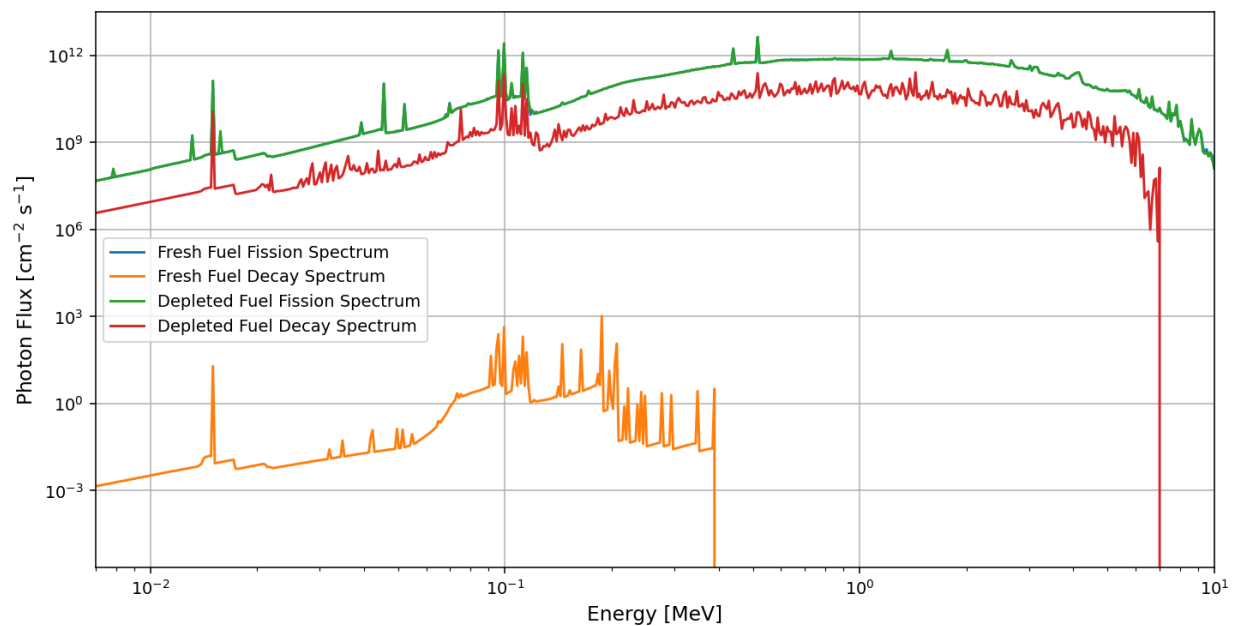


Figure 5-8: MSR gamma spectra with different fuel compositions (log y-axis). The two fission spectra are of the same magnitude and appear overlapped.

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Figure 5-8 also compares the gamma decay spectra at these points in time. Similar to the neutron spectra in Figure 5-7, gamma features, such as peaks, between the different spectra occur because of the composition of the salt and MSR structure. Figures 5-7 and 5-8 indicate that the gamma signatures will be stronger than the neutron signatures after the reactor shuts down, as expected.

Overall, the neutron and gamma fission spectra do not change significantly as a function of burnup. The neutron and gamma decay spectra show a significant increase in flux and activity because of irradiated materials.

5.4.1 Example Spent Fuel Decay Neutron Calculation

To determine the origin of decay neutrons produced by spent fuel, SOURCES-4C was used to calculate the source term for the MSR fuel salt using mass compositions from Tables 5-1 and 5-3. Neutron source terms are calculated over the 100 days of burnup to determine trends and applicability to safeguards measurements. Percentage and absolute source strengths of (α, n) and spontaneous fission (SF) neutron contributions with respect to time are shown in Figures 5-9 and 5-10. The normalized energy spectra of the (α, n) and SF neutron contributions at 100 days is shown in Figure 5-11.

Fresh fuel decay neutrons were dominated by SF, as both ^{235}U and ^{238}U have long half-lives, resulting in little alpha emission and thus a low number of (α, n) neutrons. However, shortly after operation, as other transuranics with shorter alpha half-lives are produced by neutron absorption, the (α, n) neutrons becomes the dominant source of neutrons in the fuel salt because of the high concentration of the target atoms ^{37}Cl and ^{23}Na in the salt.

In addition, the energy spectra show that the (α, n) neutron spectrum involves neutrons less than 2 MeV and emitted SF neutrons will emit at higher energies over a wider spectrum. These energy spectra can be converted into source definition cards in MCNP to perform simulated safeguards measurements to identify potential signatures for safeguarding the MSR, as shown in Figure 5-7 in the previous section.

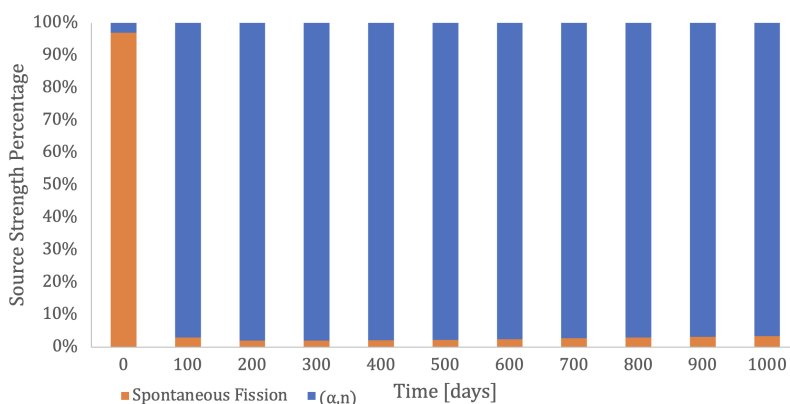


Figure 5-9: MSR (α, n) and SF neutrons percentages

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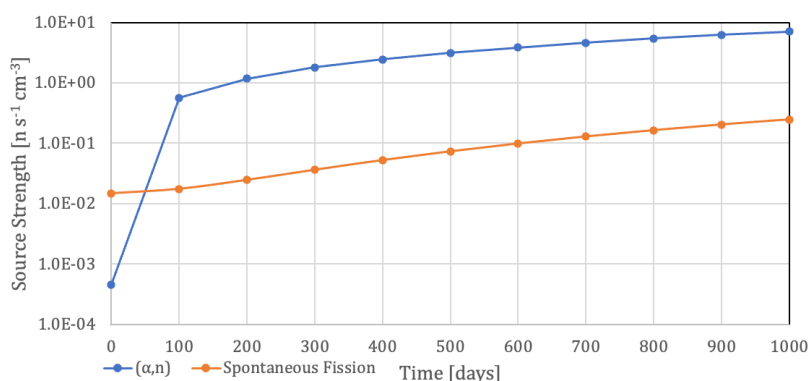


Figure 5-10: MSR (α, n) and SF neutron source strength

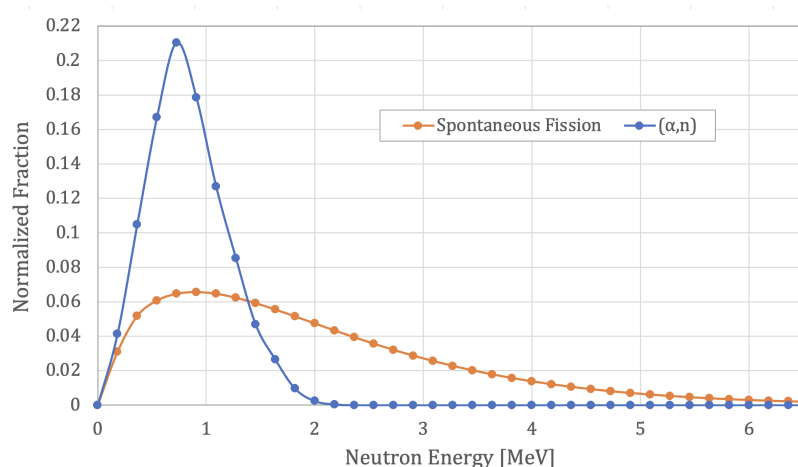


Figure 5-11: MSR neutron spectra of (α, n) and SF neutrons at 100 days

SOURCES-4C is not just applicable to MSR salt but also can be applied to any spent fuel given the burnup isotopics of actinides and low-Z target materials. Solid reactor fuels are expected to have different decay neutron source terms as there would not be as high a concentration of (α, n) target materials as there are in molten salts.

This analysis is presented for the MSR as a proof-of-concept. Although not provided here, a similar approach could be taken for the other reactor designs to determine the production mechanism for the gamma and neutron signatures.

5.5 Molten Salt Reactor MC&A Considerations

The MSR design simulated for this study presented several MC&A challenges. The design is fundamentally different from a traditional LWR, and is further complicated by the liquid form of the fuel. To highlight some of these challenges and additional considerations, some MC&A relevant parameters and examples are provided.

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The material quantities and isotopics provided in Table 5-3 were used as the basis of the example calculations for the simulated MSR.

5.5.1 Molten Salt Reactor SNM Quantities

Liquid fuel does not lend itself to item counting, so for the purposes of these discussions, this reactor design is considered a bulk facility. It should be noted this may not be the case for areas outside of the reactor core that contain nuclear material (e.g., shipping or receiving locations), but a facility level assessment is beyond the scope of this effort.

The reactor design simulated for this project includes feed and removal of fission products; thus, depending on the time frame under consideration, material could be added to the system.

The values in Section 5.2 show that, as expected, the uranium and plutonium inventories change over time. To provide an example of the magnitude of change expected for a specific time frame, uranium and plutonium quantities were compared over a period of 300 days. For this example, the beginning values were taken from the first time step, and the final uranium and plutonium values were taken from the thirtieth time step. Over this period there were 30 additions amounting to 360 kg of HALEU, and, as stated previously, it is assumed that no SNM was removed. Based on these assumptions, the calculated changes in uranium and plutonium are

$$\begin{aligned}\text{Change in uranium} &= 880 \text{ kg} \\ \text{Change in plutonium} &= -300 \text{ kg}\end{aligned}$$

Note that a positive value indicates there is less material present at day 300 than at day 0, and a negative value indicates there is more material present at day 300 than at day 0. The change in uranium and plutonium over each of the time frames tracked in Table 5-3 illustrates that uranium depletion and plutonium production follow approximately linear trends (Figure 5-2). However, due to the feed of fresh fuel, some fluctuations are expected, which will necessitate a comprehensive understanding of the expected values and accurate SNM measurements to determine if fluctuations are outside of the expected material changes.

An important MC&A consideration is that for traditional facilities it is assumed the SNM can be tracked throughout a process or within a particular containment area. This assumption is reasonable for item facilities or bulk facilities where the inputs and outputs for different stages can be quantified to verify that the total amount of SNM is still present. However, in this simulated MSR example, the amount of SNM present is changing as a function of time in the reactor system. The change is expected, given the nature of reactor operation, but the amount and location of SNM present will be dynamic.

5.5.2 Molten Salt Reactor SNM Quantification Uncertainty Considerations

An important consideration for tracking material quantities is the uncertainty of any measured value. To illustrate the impact of uncertainty on tracking changes in SNM, an example calculation is provided for the predicted change in uranium and plutonium. For this example, the uncertainty associated with each term used to calculate the change in uranium and plutonium must be known. Multiple measurements, or calculations, may be required to determine each individual term, which

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would require the error to be propagated appropriately for every value prior to determining the total uncertainty. Also, due to the nature of the measurement scenario, in calculations for a physical MSR, the values and their associated uncertainties may not be independent. For example, if there are several material balance periods and the initial and final values are not independent, then the covariance would also need to be included when calculating the total uncertainty. To fully address these considerations, a thorough analysis of the measurement techniques and the measurement points that could be used would be required. Detailed simulations could be used to generate the signature predictions and a simulation tool would likely be required to assess the overall uncertainty.

To provide some preliminary parameters for assessing the uncertainty associated with the predicted changes in uranium and plutonium for this reactor design, a simplified calculation is performed. The following assumptions were made for this example:

1. The (total) mass of uranium is measured with 1 percent uncertainty.
2. The (total) mass of plutonium is measured with 1 percent uncertainty.
3. No uranium or plutonium is removed.
4. Every 10 days, 12 kg of HALEU uranium is added (as described in Section 5.2).
5. Additions were measured with 1 percent uncertainty.
6. No correlated uncertainties are considered.

Based on these assumptions, the approximate uncertainties associated with the previously calculated changes in uranium and plutonium mass are

$$\text{Uncertainty}_U = 1,169 \text{ kg}$$

$$\text{Uncertainty}_{Pu} = 10 \text{ kg}$$

These uncertainties were calculated using basic error propagation methods. Although current reactors do not measure uncertainty, it is informative to provide some context in terms of MC&A parameters. These values show that a 3σ confidence interval would be approximately 78,893 to 85,907 kg U, or 7,014 kg U, which is approximately 9 percent of the ending inventory. And for Pu, the 3σ confidence interval would be 959 to 1,018 kg Pu, or 59 kg Pu, which is approximately 6 percent of the ending inventory.

Different approaches could be explored to lower the uncertainty. Some of these options include different measurement frequencies and selecting measurement techniques with lower uncertainties. However, considerations related to these options would need to be assessed. For example, more frequent inventory intervals would lower the associated uncertainty; however, a frequent static physical inventory for an MSR will be challenging. Static physical inventories that require the salt to be completely drained could be possible periodically when maintenance is performed, but this schedule is likely to be on the order of years, not months [23].

Further, this example scenario did not consider the uncertainty associated with any removals due to operation. This would increase the values presented here, and if the term was based on a measurement of the existing inventory, then the correlations between terms would need to be included.

5.5.3 Molten Salt Reactor Safeguards Discussion

The amount of SNM that will be present in an MSR, combined with the liquid form of the fuel, will make direct material tracking challenging, and could require novel approaches for material quantification. It is uncertain if the current physical inventory frequencies will be applicable for these reactor types. If a static physical inventory for an MSR requires taking the reactor offline every nine months (as per category II fuel cycle facility), that would be accompanied by a significant cost.

The quantity of material that would need to be tracked, combined with measurement challenges (e.g., material access, hostile measurement environments, uncertainty requirements), are some of the factors driving the consideration of alternative approaches to physical inventories. The use of multiple control areas (CAs) is one option, which is also noted by other researchers [24].

If the CA structure is configured such that the reactor is one CA, then what enters and exits the reactor would need to be tracked. In this example, the amount and isotopic composition of any material entering or leaving the reactor MBA can be measured. Note that there could be different options for what units of material are tracked prior to entry into the reactor, for example, bulk material, drums, etc. This could include parameters such as mass, volume, isotopics, etc. For both the entry and exit measurements, the details would be dictated by the specifics of the reactor design (i.e., online feed and removal). One of the key aspects to this approach is that the SNM leaving the reactor CA can be correlated to the expected quantities for a given set of reactor operating parameters.

Models and simulations could provide a link to relate material composition and reactor operating parameters. If it can be demonstrated that simulations can predict the expected material values under different conditions, then what enters the reactor can be tracked and compared to measurements of what is removed. Further research is required on the exact measurements that would need to be made and the reactor parameters that would need to be monitored.

6 Gas-Cooled Reactor

The GCR geometry modeled for this study is adapted from the neutronic benchmark created by Connolly et al. [25]. The model uses 10 percent enriched TRISO fuel and is depleted over 2 years at 250 MWth. The prismatic core is comprised of hexagonal graphite blocks. Each block contains helium channels and TRISO fuel compacts. Some blocks contain larger channels for shutdown rods. Top-down, the fueled blocks are arranged in a ring pattern with shutdown channels on the outside.

6.1 TRISO Fuel

Both the GCR and microreactor designs in this work use TRISO fuel (Figure 6-1), which consists of a fuel kernel surrounded by four layers of encapsulation. The fuel kernels are uranium oxycarbide (UCO) for this reactor configuration. The first layer is made of porous carbon, which provides space for fission gasses to accumulate, alleviating fuel swelling. The second layer is dense, highly isotropic, inner pyrolytic carbon (IPyC), and the third is silicon carbide (SiC). The IPyC layer serves as a seal to protect the kernel from the main byproduct produced during SiC application. The outer pyrolytic carbon (OPyC) layer serves as a final barrier to fission product gasses and acts as a bonding surface.

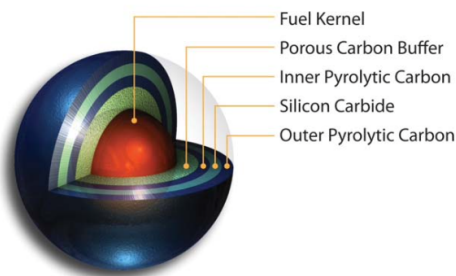


Figure 6-1: TRISO particle illustration showing the different layers [26]

TRISO fuel in the GCR geometry is sized according to the reference benchmark [25], while microreactor sizes were obtained from the U.S. Department of Energy Advanced Gas Reactor (AGR) Fuel Development and Qualification Program [26]. The TRISO dimensions for both systems are similar, with layer radii varying less than 2 percent.

6.2 Geometry

A cross section of the GCR is shown in Figure 6-2. The gray and dark green blocks represent graphite blocks, and yellow circles are coolant channels. The lighter green blocks represent TRISO compacts. Detailed cross-sections of a fuel block and a fuel compact are shown in Figure 6-3.

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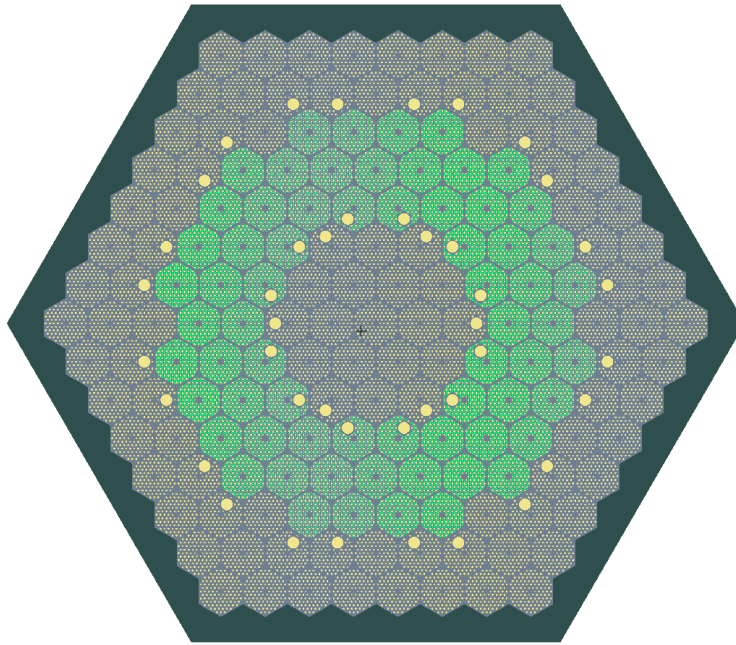
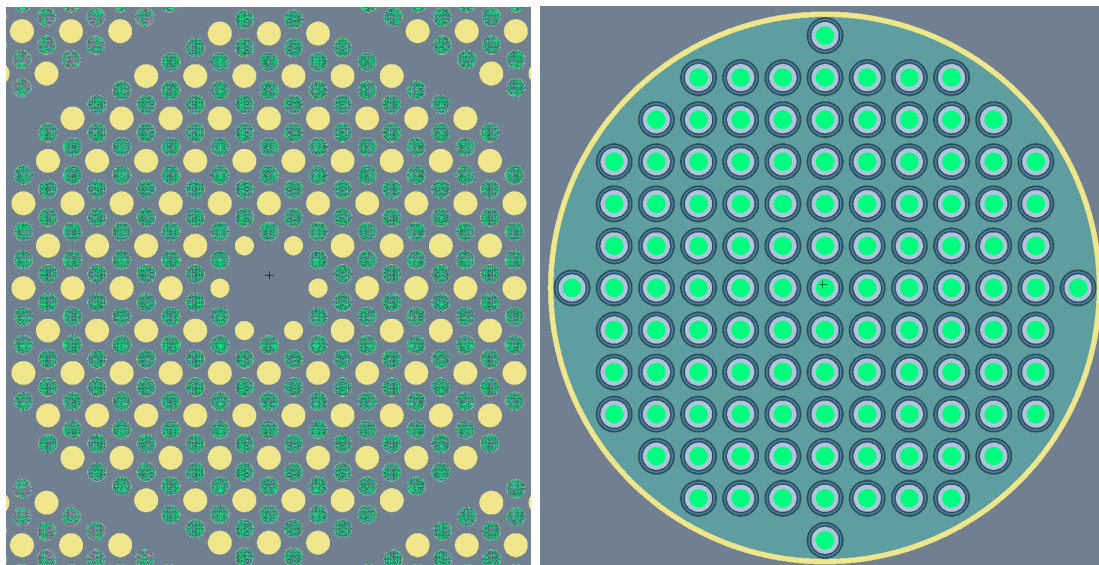


Figure 6-2: GCR radial slice. The dark green and gray are graphite blocks, the yellow represents coolant channels, and the lighter green are fuel blocks.



(a) Fuel block: gray region is graphite, yellow circles are helium coolant channels, and green circles are fuel channels

(b) Fuel compact: bright green circles are UCO kernels, sea green shading with yellow border represents graphite matrix

Figure 6-3: GCR block and fuel compact radial cross-section views

Gas-Cooled Reactor

The apothem of the core is 259.8 cm. Each layer of the core contains 169 graphite blocks, 66 of which contain fuel. Each block has an apothem of 18 cm and a height of 79.3 cm. There are 14 layers of blocks stacked vertically, resulting in a total core height of 1110.2 cm.

Each standard fuel block contains 210 fuel channels, while shutdown blocks have 186 fuel channels. Fuel compacts, which are inserted into the fuel channels, contain 35 percent by volume UCO TRISO enriched to 10 percent. Fuel blocks have 2.69 cm graphite plugs above and below the fuel compacts. Unfueled graphite blocks surround the core to act as a neutron reflector.

6.2.1 Model Assumptions

Like the benchmark [25], the model used does not have containment, shielding, or reactivity control. To simplify and adapt the problem for depletion calculations, the uniform burnable absorber is omitted and the fuel enrichment is lowered to 10 percent.

As a consequence of removing the burnable absorber, k_{eff} is significantly higher than in other models. The burnable absorber could be evaluated in future work, where several unfueled graphite compacts dispersed throughout the core will have neutron poisons such as boron carbide.

6.3 Burnup

Depletion results are not skewed by the large neutron multiplication factor, as k_{eff} is not a parameter used in the calculation. The normalized flux produced by MCNP is scaled up using reactor power (Equation 1) and a constant flux is used across the entire time step. The GCR is depleted at 250 MWth for 2 years using eight time steps. Figure 6-4 plots the GCR effective multiplication factor over time.

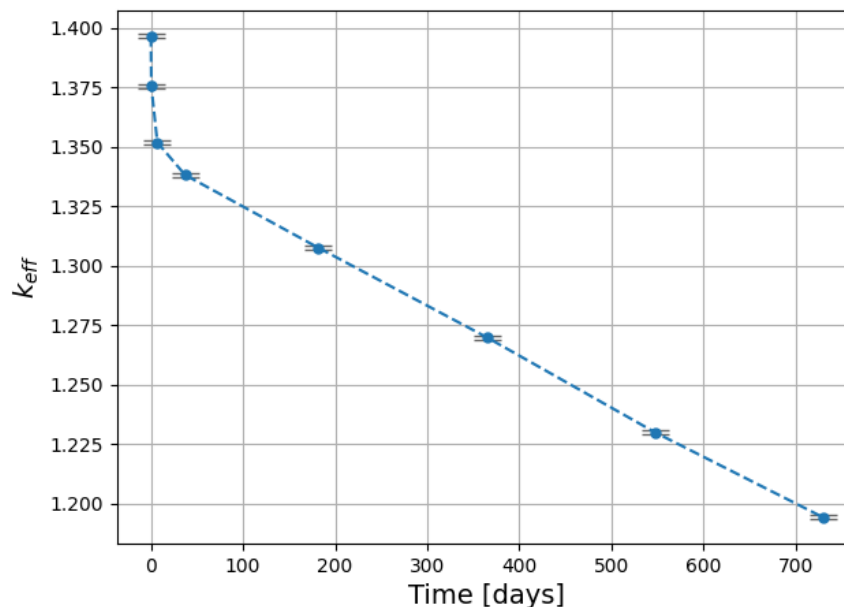


Figure 6-4: GCR k_{eff} over time

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Without a burnable absorber, the neutron flux does not change much with time, allowing for fewer time steps. However, extra steps are prioritized at the beginning of life, as the introduction of fission products to fresh fuel has a large impact on flux. This can be seen in Figure 6-4, in which the sudden drop in reactivity occurs as strong neutron absorbers (e.g., ^{135}Xe and ^{149}Sm) are created from fission events. The error bars in Figure 6-4 represent the statistical uncertainty discussed in Section 3.1.

Figure 6-5 and Table 6-1 show the mass of SNM over time, where SNM has been defined as all isotopes of uranium and plutonium. Error bars are not included in plots of nuclide mass because the propagation of error through the depletion calculation is an area of ongoing research [27].

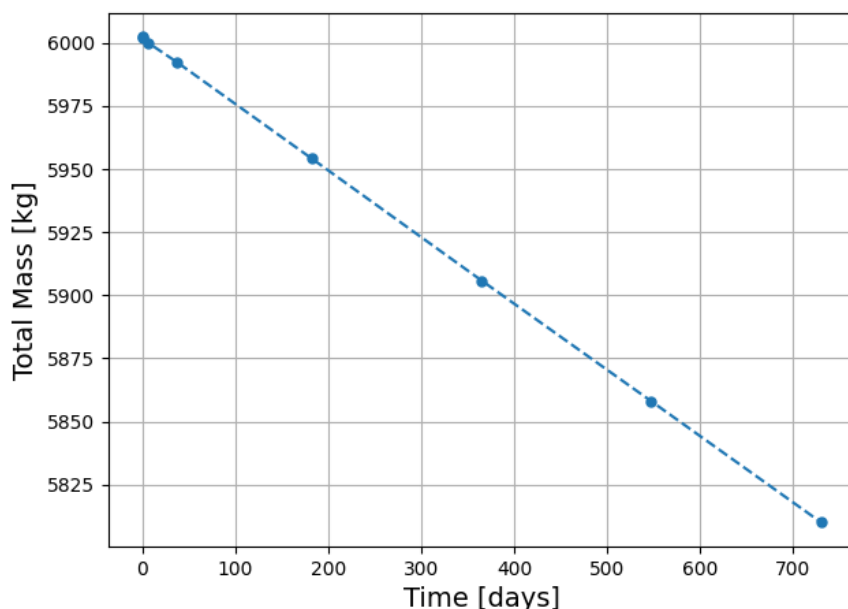


Figure 6-5: GCR special nuclear material mass over time

Table 6-1: GCR select actinide mass as a function of burnup

Time [days]	^{234}U [g]	^{235}U [g]	^{236}U [g]	^{238}U [g]	^{237}Np [g]	^{238}Pu [g]	^{239}Pu [g]	^{240}Pu [g]	^{241}Pu [g]
0	3.546e3	5.929e5	0.000e0	5.406e6	0.000e0	0.000e0	0.000e0	0.000e0	0.000e0
0.5	3.545e3	5.927e5	2.778e1	5.406e6	0.000e0	0.000e0	2.488e0	1.690e-3	0.000e0
7	3.541e3	5.907e5	3.903e2	5.405e6	7.700e-2	0.000e0	3.152e2	7.39e-1	1.690e-3
37	3.523e3	5.813e5	2.053e3	5.403e6	2.451e0	7.21e-3	2.609e3	2.622e1	6.580e-1
182	3.434e3	5.370e5	9.803e3	5.391e6	6.234e1	1.246e0	1.241e4	5.649e2	6.915e1
365	3.321e3	4.837e5	1.900e4	5.375e6	2.401e2	9.973e0	2.243e4	1.902e3	4.387e2
547	3.206e3	4.332e5	2.754e4	5.359e6	5.178e2	3.313e1	3.023e4	3.620e3	1.173e3
730	3.089e3	3.850e5	3.552e4	5.342e6	8.855e2	7.831e1	3.628e4	5.518e3	2.223e3

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Similar to temporal resolution, sufficient spatial resolution is necessary to capture the accumulation of nuclides. Isotopes produced by successive interactions are particularly sensitive to spatial resolution (e.g., ^{238}Pu , ^{241}Pu , and ^{242}Pu) [11]. The GCR model uses three radial depletion rings and fourteen axial depletion slices for a total of forty-two depletion zones. This resolution is sufficient for system values (k_{eff} and total mass) but more regions could be needed for local parameters, such as individual fuel element composition.

6.4 Uncertainties

Table 6-2 shows the Monte Carlo and nuclear data standard deviations (σ). Figure 6-6 plots the top five contributors to nuclear data uncertainties with respect to specific nuclides.

Table 6-2: GCR burnup, k_{eff} , uncertainties, and percentage of neutron energies causing fission

Time [days]	Burnup [GWd/MTU]	k_{eff}	Statistical $k_{eff} \sigma$	Nuclear Data $k_{eff} \sigma$	% thermal fissions	% epitherm. fissions	% fast fissions
0	0.00	1.314641	0.000179	0.00348	88.45%	10.81%	0.73%
0.5	0.02	1.295476	0.000195	0.00360	88.30%	10.96%	0.74%
7	0.29	1.273910	0.000172	0.00383	88.13%	11.12%	0.75%
37	1.54	1.262316	0.000177	0.00365	88.15%	11.09%	0.76%
182	7.58	1.236968	0.000181	0.00377	88.48%	10.75%	0.76%
365	15.21	1.204407	0.000176	0.00416	88.91%	10.32%	0.77%
547	22.79	1.175820	0.000178	0.00394	89.30%	9.93%	0.77%
730	30.41	1.167593	0.000168	0.00407	89.51%	9.72%	0.77%

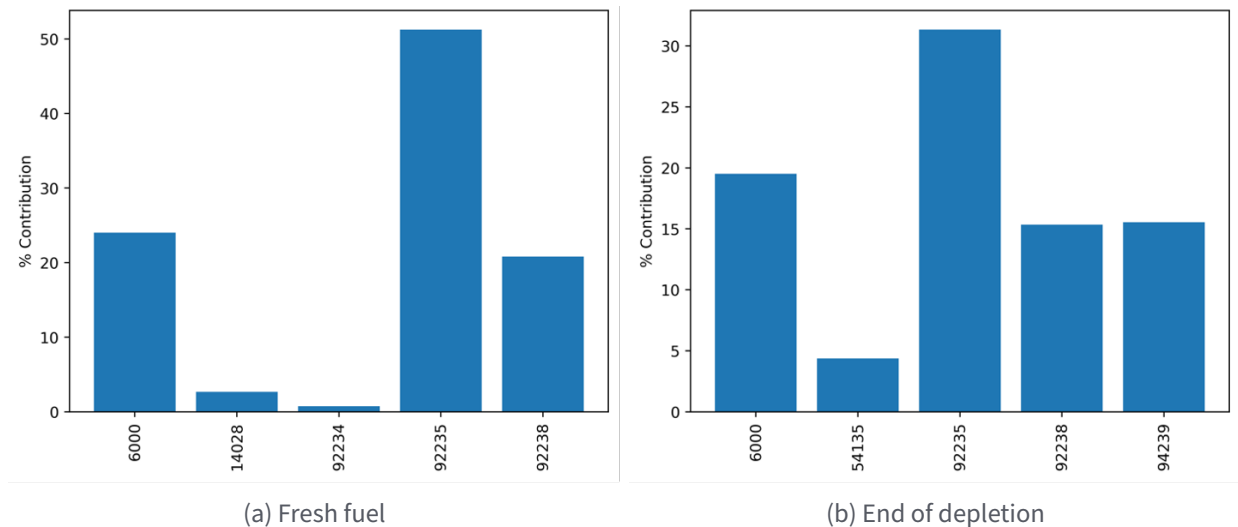


Figure 6-6: GCR top five isotopes contributing to nuclear data-induced uncertainty in k_{eff}

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As shown in Figure 6-6, ^{239}Pu (ZAID 94239) begins to play a larger role in the nuclear data-induced uncertainty at the reactor end of life (EOL). Xenon-135 is a potent neutron absorber that reaches an equilibrium concentration after several hours of reactor operation. The large absorption cross-section has a significant impact on neutron economy, and the accumulation of ^{135}Xe (ZAID 54135 in Figure 6-6) accounts for nearly 5 percent of the nuclear data uncertainty at the end of the GCR depletion simulation. Additionally, the uncertainty in k_{eff} due to ^{235}U fell from 50 percent to 30 percent during the depletion simulation. Because the GCR contains a lower enrichment than other concepts, a higher fraction of ^{235}U is fissioned or transmuted into other nuclides.

The average statistical uncertainty in the MCNP calculation of k_{eff} is 0.00018 (18 pcm). The average uncertainty in k_{eff} resulting from nuclear data is 0.0038 (379 pcm). The larger fraction of thermal neutrons in the GCR results in a lower nuclear data-induced uncertainty compared to the MSR, which has more epithermal neutrons. The resonance region in the epithermal (i.e., intermediate) neutron energy range typically has a much greater uncertainty than the thermal and fast regions. The top twenty nuclide interaction uncertainty contributors to k_{eff} are shown in Figure 6-7.

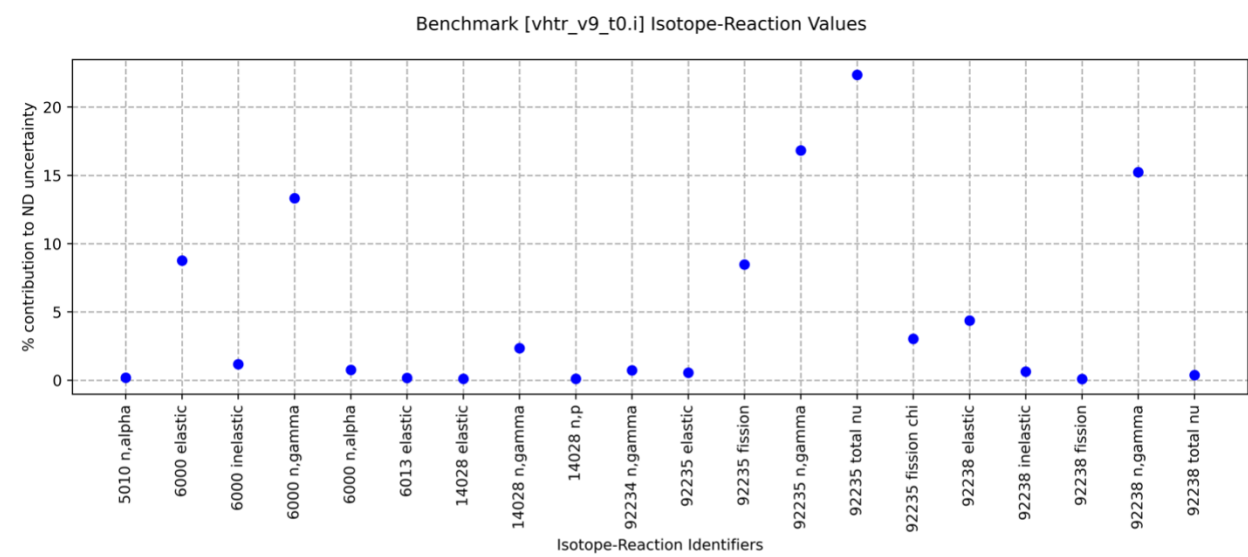


Figure 6-7: GCR top twenty nuclide interaction uncertainty contributors to k_{eff} with fresh fuel

Both the ^{235}U and ^{238}U (n, γ) cross-section uncertainties make up a large percentage of the total nuclear data uncertainty for k_{eff} . The GCR concept has the largest fraction ^{238}U (n, γ) cross-section uncertainty compared to the MSR and microreactor concepts due to the lower ^{235}U enrichment level in the GCR.

6.5 Neutron and Gamma Energy Spectra

The neutron and gamma energy spectra for the GCR design were simulated to examine changes for different fuel compositions. Figure 6-8 shows the fresh and depleted fuel neutron fission spectra.

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The flux profiles are similar over most of the energy range; to identify changes in the flux spectra, the depleted fuel percent difference from fresh fuel is shown in Figure 6-9.

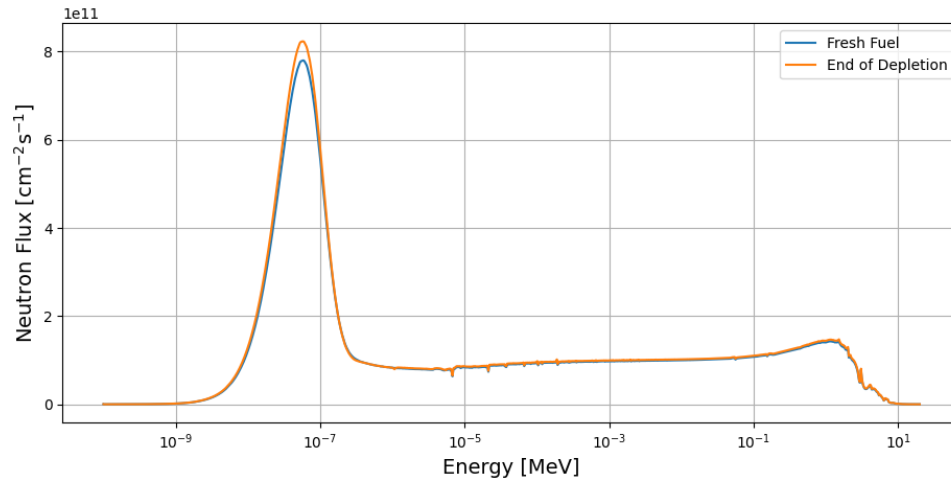


Figure 6-8: GCR neutron spectra with different fuel compositions

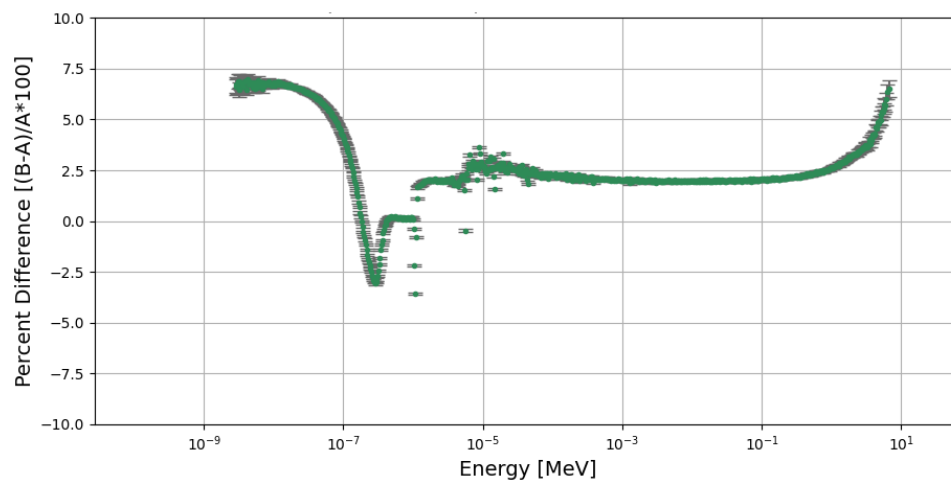


Figure 6-9: GCR neutron spectrum: depleted fuel percent difference from fresh fuel

The large amount of graphite in the GCR produces a thermal neutron spectrum. The flux is generally higher at the end of depletion because plutonium produces more neutrons per fission than uranium for an equivalent reactor power. It should be noted that differences resulting from intensities less than 0.01 percent of the total are not plotted, as small neutron fluxes have large standard deviations. These differences occurred on the left and right extremes of Figure 6-9. The statistical uncertainty for each tally is propagated through the percent difference equation to calculate the

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error bars shown in Figure 6-9. The error bars represent the 95 percent confidence interval.

The gamma spectra for fresh and depleted fuel are shown in Figure 6-10. The gamma spectrum produced from neutron interactions is generated in a separate simulation from those produced by fission product decay. During operation, the gamma spectrum would be the combination of the fission and decay spectra. If fuel were removed from the reactor and measured, just the decay spectrum would be observed. An alternate version of Figure 6-10 with a linear y-axis can be found in Appendix A.

The fresh fuel decay spectrum has a low total intensity compared to other spectra because fresh fuel does not have any fission products and uranium has a long half-life. Differences in the fission spectrum (red and yellow lines) are small and difficult to visually identify using a log y-axis, but are more easily observed using a linear scale (Figure A-1). The decay gammas generated from spent fuel are an order of magnitude less intense than those generated during reactor operation. The contribution to the gamma spectrum from decay is negligible during operation, but important to consider if fuel is measured while the reactor is shutdown since this would be the prominent signature.

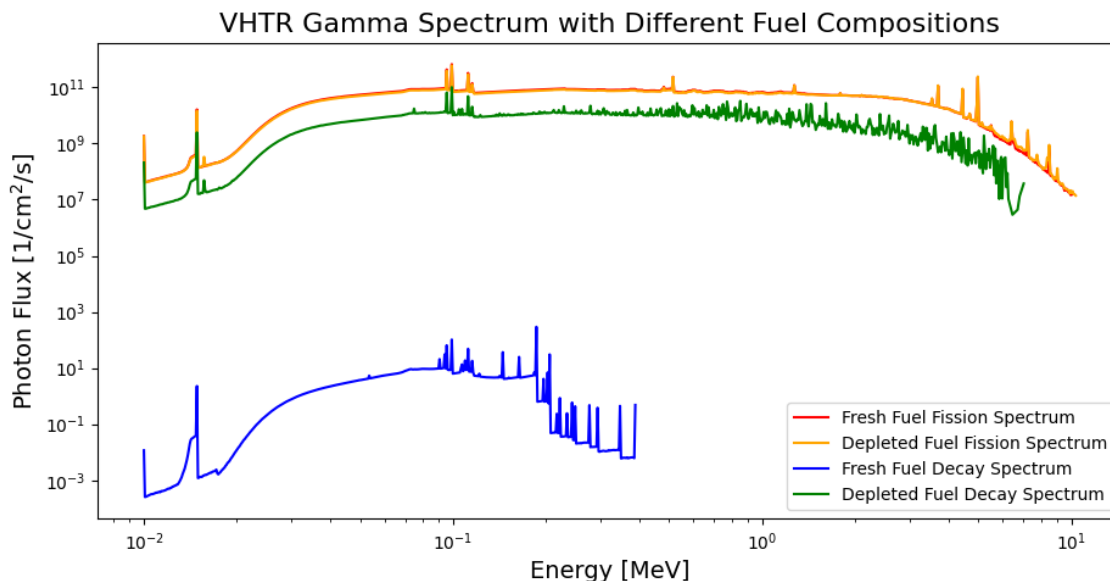


Figure 6-10: GCR gamma spectra with different fuel compositions (log y-axis). The two fission spectra are of the same magnitude and appear overlapped.

The origin of the contributions to the gamma and neutron flux are important considerations when developing and assessing MC&A approaches. A more detailed analysis of the different signatures should be performed in future work.

6.6 Gas-Cooled Reactor MC&A Considerations

The GCR design modeled for this project contained approximately 6000 kg of uranium and no plutonium at the beginning of life (BOL). As discussed in Section 4, the application of material

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category-specific MC&A requirements is not established. But to provide context for the scenarios and examples provided here, Category I requirements (selected based on the total amount of SNM) were used as a basis for the discussion points. The calculations and examples for this reactor design were based on the geometry and burnup information provided in Sections 6.2 and 6.3. Recall that the core contains 66 stacks of 14 blocks each (a block is shown in 6-11). It is assumed, based on geometry calculations, that each block contains 6300 compacts (the shutdown blocks will have approximately 5580 compacts), for a total of 5,821,200 compacts in the core. Note that the quantity and location of the TRISO compacts remains static during operation.

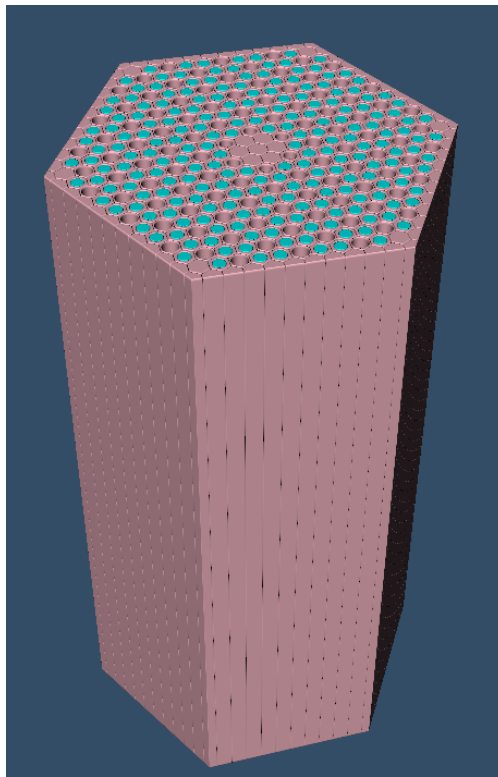


Figure 6-11: 3D rendering of single GCR fuel block. Pink is graphite and blue is TRISO compact reflector plugs.

6.6.1 Gas-Cooled Reactor SNM Quantities

The changes in SNM quantities for this design, as with a traditional reactor design, are due to isotopic changes during the course of operation (both production and consumption), and not to changes in the number of fuel elements. The amount of uranium at the BOL, as previously stated, will be approximately 6000 kg (and no plutonium will be present). At the EOL, there will be approximately 5800 kg of uranium and 44 kg of plutonium. Thus, the expected change due solely to operation will be approximately 200 kg for uranium and -44 kg for plutonium.

In addition to the total amount of material in the core, the SNM contained in different units (e.g., compacts, blocks, stacks) is also relevant for MC&A considerations. If it is assumed that all

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compacts have the same initial isotopic composition, then any changes in SNM can be calculated as a function of burnup region within the core. For one of the blocks in the highest burnup region, the SNM present in an individual compact at the BOL is approximately 1.05 g uranium (no plutonium), and at the end of the simulation period approximately 0.999 g uranium and 9 mg plutonium. Thus, an individual regular block in this region will contain approximately 6.6 kg of uranium and no plutonium at the BOL. At the EOL, the same block will contain 6.3 kg uranium and 0.06 kg plutonium.

The amount of material in a stack can be computed based on the values for an individual block. If it is assumed that all blocks in a single stack have the same burnup, and there are fourteen blocks per stack, then for the high burnup location there will be approximately 93 kg of uranium per stack at the BOL. At the EOL, there will be approximately 88 kg of uranium and 0.9 kg of plutonium per stack. These values, and some additional parameters, are summarized in Table 6-3. Note that the values in this table are based off geometrical considerations and from one material region in the core (which is the region with the highest burnup). The SNM quantities, which are approximations that are meant to provide context for the MC&A considerations, were calculated using the final simulation results (i.e., EOL).

Additional considerations related to these parameters are discussed in 6.6.3.

Table 6-3: Approximate SNM quantities for different units in the GCR based on the values for the end of the simulation period.
Note, the block and stack values are assuming a regular block configuration, not a shutdown block.

Unit	No. of Compacts	Uranium (g)	Plutonium (g)
Compact	1	1	0.01
Fuel Channel	30	30	0.29
Block	6300	6290	62
Stack	88,200	88,100	862
Total	5.7M	5.8M	44,000

6.6.2 Gas-Cooled Reactor SNM Quantification Uncertainty Considerations

The uncertainty associated with the SNM quantification depends on the measurements and parameters that are used for the inventory. The relevance of this metric depends on the MC&A procedures that are implemented. For example, if a measurement such as gamma ray spectroscopy is performed on every compact, then the measurement error associated with this method will need to be propagated. However, if the inventory is based on counting the total number of compacts present, then no change is expected, and any inventory difference would be due to counting errors. To illustrate the potential impact of counting errors, consider a scenario where 1 percent of the compacts are miscounted. In that situation, approximately 57,000 compacts would be miscounted, which, for the high burnup region considered for the example in the previous section, would result in an inventory discrepancy of approximately 58 kg of uranium. So, for situations where the inventory relies upon counting, the required controls will need to be considered.

The error that would be associated with individual compact measurements could also be exam-

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ined. If it is assumed that each compact will be measured, and the uranium mass determined to within 1 percent, the final uranium mass will have a range of approximately 116 kg. Thus, even if the number of items does not change, there will still be significant uncertainty associated with this uranium declaration.

As an alternative approach, the inventory of larger units, for example, blocks, could be considered. This concept is further discussed in Section 6.6.3.

6.6.3 Gas-Cooled Reactor Safeguards Discussion

The GCR design consists of solid fuel blocks, unlike the liquid-fueled MSR design. Due to this fuel form distinction, different MC&A metrics and considerations apply. As with traditional LWR configurations, there are set units of nuclear material, but in a different geometry (i.e., not in traditional LWR fuel assemblies).

As was noted in Section 6.6, a large number of TRISO compacts comprise the total fuel for this reactor design (approximately 5.7 million). Thus, tracking compacts on an individual basis may not be feasible in all scenarios. However, other units could be considered, e.g., if blocks or stacks were tracked, this would reduce the number of items. In the configuration simulated for this study, there were 924 blocks in 66 stacks. A method to contain the blocks or the stacks (similar to fuel rods and assemblies) could provide a reduced number of items to be tracked for MC&A purposes.

Another consideration related to determining the items that would be tracked for MC&A is the associated mass of nuclear material. Individual compacts will have approximately 1 g of uranium, whereas a stack will contain approximately 90 kg. A traditional PWR fuel assembly has approximately 400 kg of uranium, so tracking a stack would have greater similarity to current MC&A approaches than accounting for individual compacts. Items are discrete units with unique identification, and it would have to be determined if a stack is a discrete unit.

It should be highlighted that in this design, the compacts were stationary during operation. Tracking by batch as an accounting unit is another tracking concept that could be considered. For MC&A purposes, this means that even if each compact has identical isotopics at the BOL, differences will be expected at EOL based on their location in the core. Thus, signatures that could be correlated to reactor operating parameters would need to be established to determine what could be used as inventory metrics (i.e. to quantify the SNM). Further, if fuel shuffling were performed the effect on material tracking would need to be considered. The statistically significant changes identified in the different signatures, and their correlation to nuclear material quantity, would also need to be established. Further modeling and simulation research may be performed to provide expected signatures and support the development of an MC&A approach until sufficient experimental and operational data is acquired.

7 Microreactor

Snowflake is a heat pipe-cooled, yttrium hydride (YH)-moderated microreactor concept developed by Los Alamos National Laboratory [28]. The number of units (i.e., sets of fuel rods, moderator cylinders, and heat pipes) in the design depends on the power level required for various mission needs. The model used in this work has six control drums for reactivity control and seven channels for graphite plus emergency shutdown rods. In this configuration, Snowflake produces 3.5 MWth over a four-year lifetime.

7.1 Geometry

A radial slice of the Snowflake microreactor is presented in Figure 7-1. The core is comprised primarily of graphite (purple). The regions surrounding the core and a majority of the control drums are made of beryllium oxide (BeO) (dark blue). Boron carbide is used for the absorption region of the control drums (orange). The first outer ring is a 3 cm-thick lithium hydride (LiH) neutron shield (yellow). The thickest outer ring is a depleted uranium gamma shield (green). The last layer is stainless steel casing (light blue). The reactor is 253 cm tall and 176 cm in diameter; the active core is a hexagonal prism 170 cm tall with a 54 cm apothem.

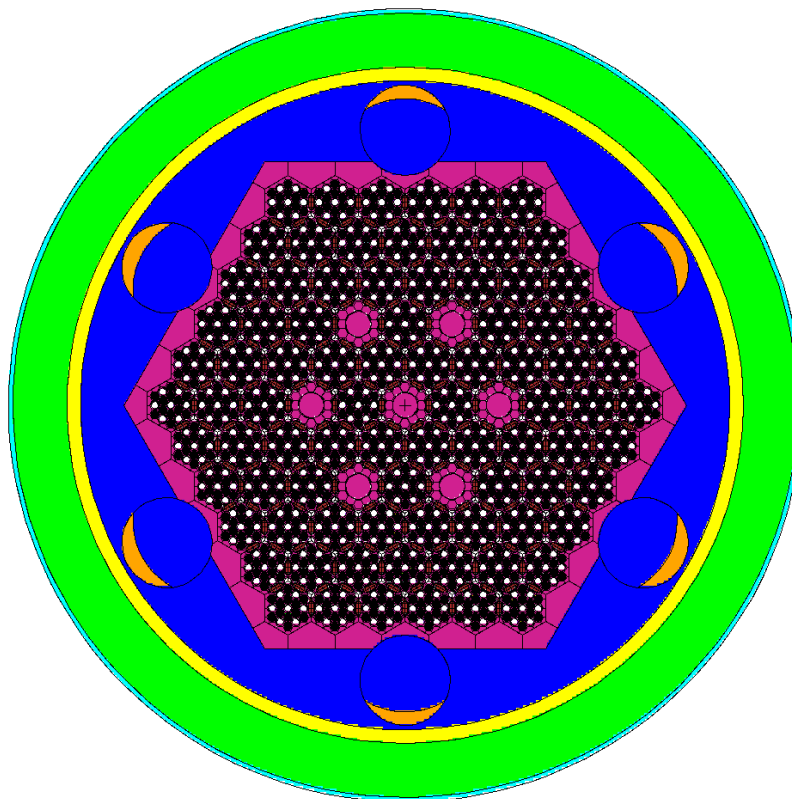


Figure 7-1: Snowflake microreactor radial slice

Microreactor

The hexagonal unit cells are known as “flakes” (Figure 7-2) and fill the Snowflake core. The stadium-shaped YH moderator (brown) surrounds each unit cell. There are 12 TRISO compacts per unit cell (black), along with 7 full heat pipes and 6 $\frac{1}{3}$ heat pipes on the corners (white). The fuel compacts are 19.5 percent enriched TRISO with a 37 percent packing fraction. Each compact is 1.7 cm in diameter.

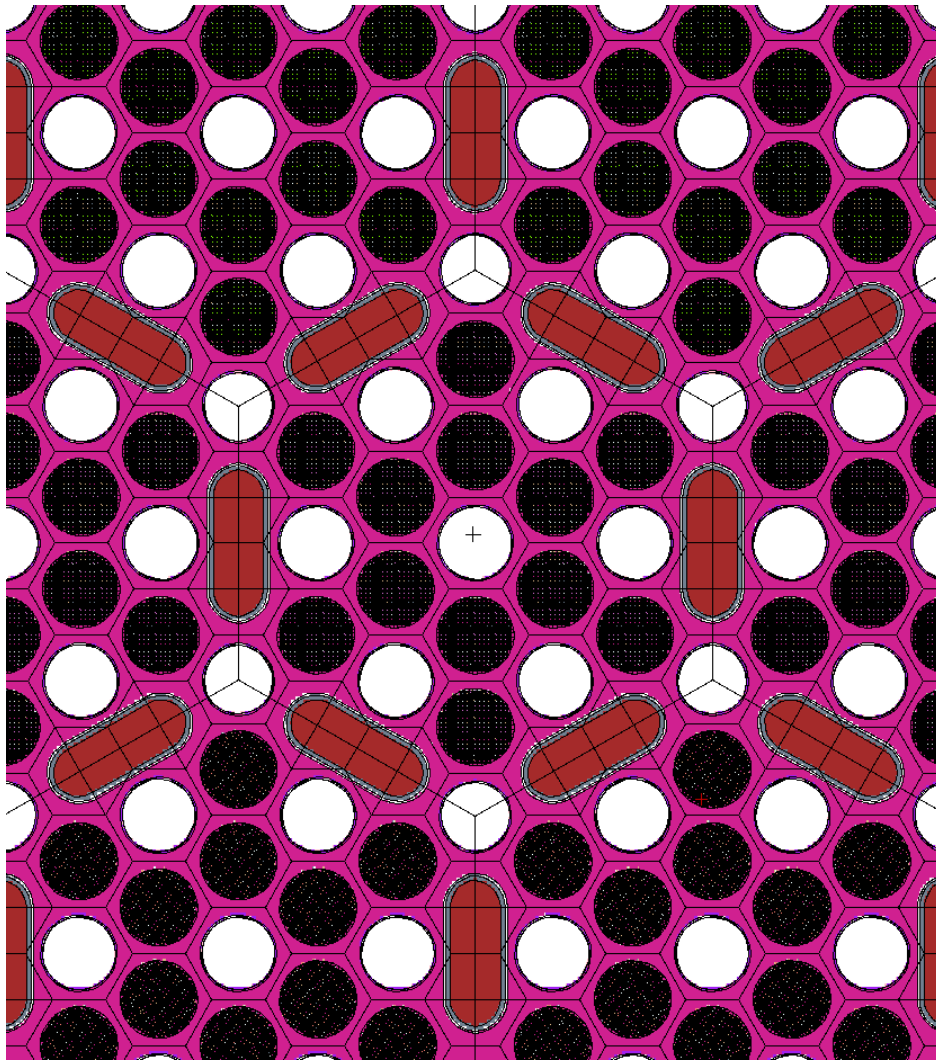


Figure 7-2: Snowflake unit cell (with partial neighbors)

7.1.1 Model Assumptions

The Snowflake microreactor used in this study is a non-optimized concept, meaning that to simplify the analysis, no burnable absorber is present to flatten the power distribution. The heat pipes were modeled using concentric cylinders representing an interior void encompassed by wick, sodium, and

clad materials. The heat pipes do not penetrate the shielding. The TRISO particles were modeled individually using a regular cubic lattice. Lastly, the control drums are fixed and do not rotate during the depletion simulation.

7.2 Burnup

The combination of three radial and nine axial depletion zones was chosen to approximate the spatial flux distribution [11]. After four years at 3.5 MWth, the peak burnup in Snowflake is 44 GWd/MTU in the center of the core. On the core periphery, burnup is much lower at 24 GWd/MTU. The average burnup is 30 GWd/MTU. The mass of SNM in the core for this design changes almost linearly with time (Figure 7-3). Table 7-1 shows select actinide masses as fuel is depleted.

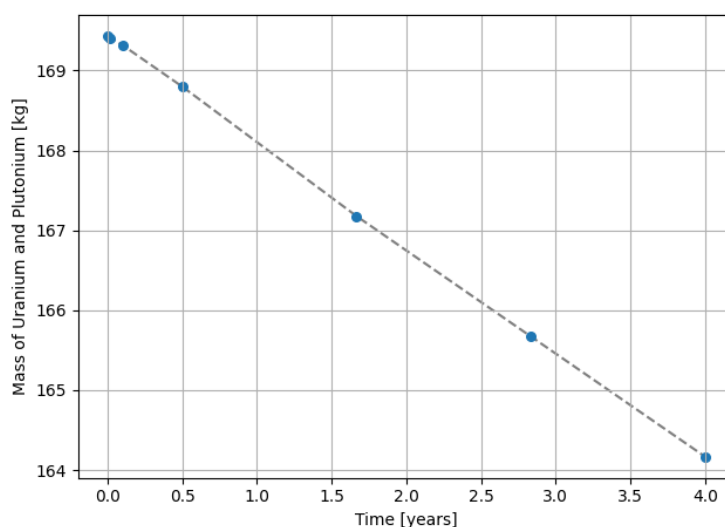


Figure 7-3: Snowflake special nuclear material over time

Table 7-1: Snowflake select actinide mass as a function of burnup

Time [days]	²³⁴ U [g]	²³⁵ U [g]	²³⁶ U [g]	²³⁸ U [g]	²³⁷ Np [g]	²³⁸ Pu [g]	²³⁹ Pu [g]	²⁴⁰ Pu [g]	²⁴¹ Pu [g]
0.0	0.000	3.303e4	0.000	1.364e5	0.000	0.000	0.000	0.000	0.000
7.0	7.435e-4	3.300e4	5.904	1.364e5	1.117e-3	0.000	3.003	2.593e-3	0.000
37.0	3.935e-3	3.286e4	3.117e1	1.364e5	2.157e-2	2.285e-7	2.491e1	1.034e-1	4.088e-4
182.0	1.923e-2	3.222e4	1.523e2	1.363e5	3.532e-1	2.187e-3	1.256e2	2.529	8.688e-2
608.0	6.229e-2	3.037e4	4.981e2	1.359e5	3.353	7.058e-2	3.829e2	2.476e1	3.159
1034.0	1.047e-1	2.858e4	8.302e2	1.356e5	9.169	3.416e-1	5.865e2	6.194e1	1.293e1
1460.0	1.487e-1	2.683e4	1.150e3	1.353e5	1.757e1	9.798e-1	7.463e2	1.074e2	3.019e1

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7.3 Uncertainties

The uncertainty on the Monte Carlo statistics of k_{eff} in MCNP is 0.00016 or 16 pcm (average of time steps). The uncertainty due to the nuclear data on k_{eff} is 0.0025 or 250 pcm (average through time steps). Table 7-2 shows the Monte Carlo and nuclear data standard deviations (σ).

Table 7-2: Microreactor burnup, k_{eff} , uncertainties, and percentage of neutron energies causing fission

Time [days]	Burnup [GWd/MTU]	k_{eff}	Statistical k_{eff} σ	Nuclear Data k_{eff} σ	% thermal fissions	% epitherm. fissions	% fast fissions
0	0.00	1.072437	0.000160	0.00239	84.29%	14.18%	1.53%
7	0.14	1.061459	0.000161	0.00245	84.15%	14.29%	1.55%
37	0.76	1.057496	0.000151	0.00247	84.14%	14.31%	1.55%
182	3.76	1.046883	0.000164	0.00249	84.19%	14.25%	1.56%
608	12.56	1.025552	0.000159	0.00244	84.56%	13.87%	1.57%
1034	21.36	1.003059	0.000156	0.00251	84.92%	13.50%	1.58%
1460	30.16	0.979013	0.000149	0.00280	85.30%	13.10%	1.60%

Figure 7-4 plots the top five nuclides that contribute to nuclear data uncertainties.

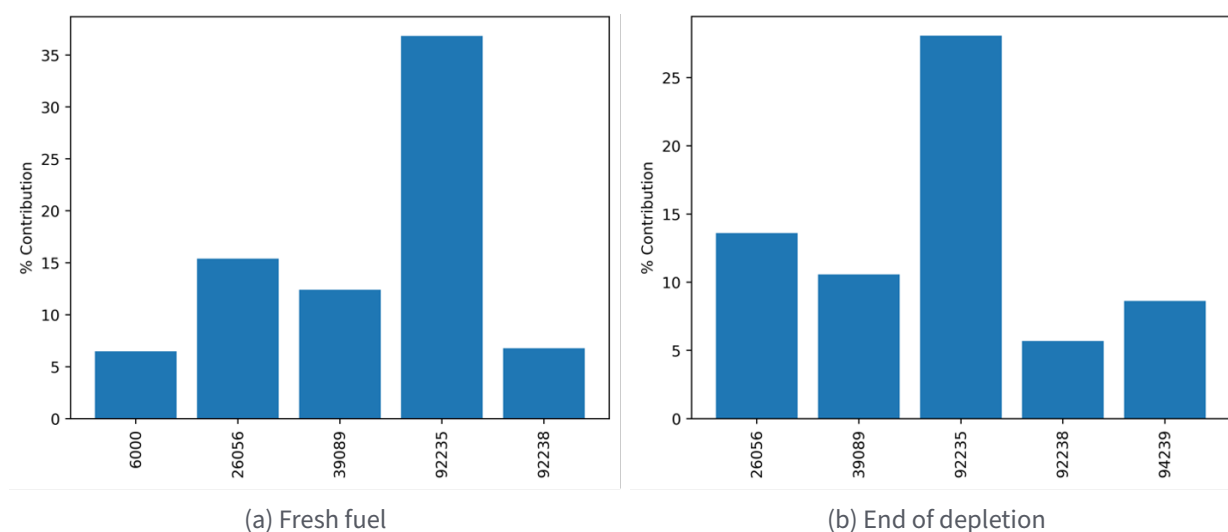


Figure 7-4: Snowflake top five isotopes contributing to nuclear data induced uncertainty in k_{eff}

The Snowflake microreactor design has a lower nuclear data-induced uncertainty compared to the other reactor systems modeled since the microreactor uses hydrogen for a significant amount of the moderation (YH). Hydrogen has well known cross-section data, so the uncertainty is lower than most nuclear data (including graphite, which the GCR relies on for moderation). In other words, the difference in uncertainty between the microreactor and the GCR is due to the materials used

in the reactor, and the difference between this microreactor or the GCR and the MSR comes from the neutron spectrum (Section 6.4).

As shown in Figure 7-4, plutonium-239 begins to play a larger role in the nuclear data-induced uncertainty as time progresses. ^{56}Y (present in the metal hydride moderator, YH_2) and ^{56}Fe (present in the heat pipes as structure) both contribute significantly to the uncertainty in k_{eff} because they are both neutron absorbers.

The top twenty nuclide interaction uncertainty contributors to k_{eff} with fresh fuel are shown in Figure 7-5.

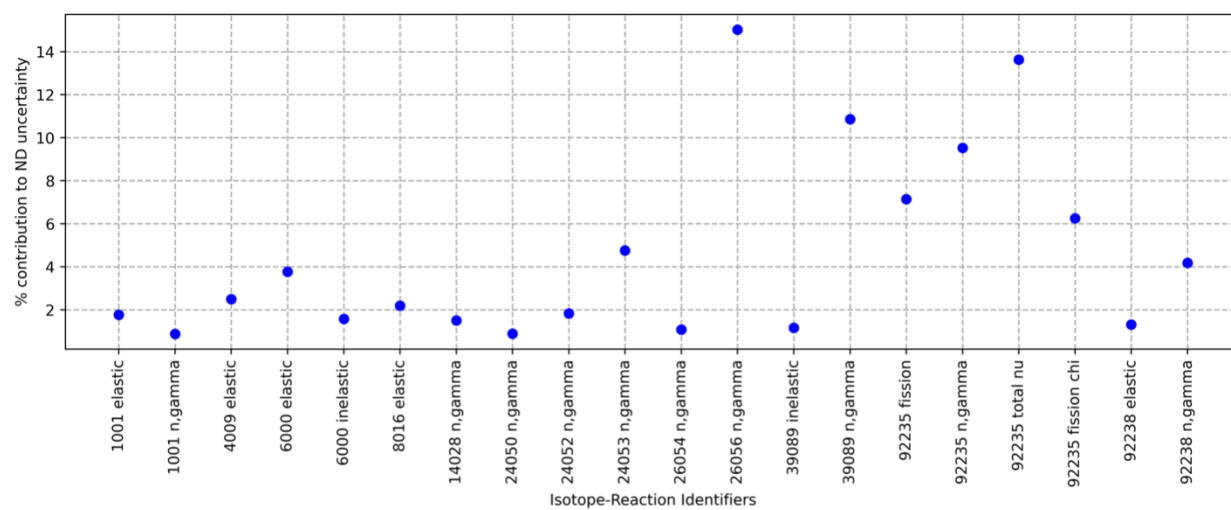


Figure 7-5: Snowflake top twenty nuclide interaction uncertainty contributors to k_{eff} with fresh fuel

Compared to the other systems, the Snowflake microreactor is unique in that it has a non- ^{235}U nuclide interaction that contributes the most to the nuclear data-induced uncertainty in k_{eff} with fresh fuel. The ^{56}Fe (n, γ) radiative capture cross-section makes up about 15 percent of the nuclear data-induced uncertainty due to the large number of heat pipes in the system.

7.4 Neutron and Gamma Spectra

The neutron spectrum in Snowflake has a much more prominent fast component than the GCR. This is also shown in the percentage of fast neutrons (energy > 100 keV) causing fission. In Snowflake, 1.53 percent of fissions are caused by fast neutrons, compared to only 0.69 percent in the GCR. This is due to the larger core size and graphite to fuel ratio in the GCR, which allows for more neutron moderation than this smaller microreactor.

The neutron spectra for fresh fuel and at the end of depletion are shown in Figure 7-6.

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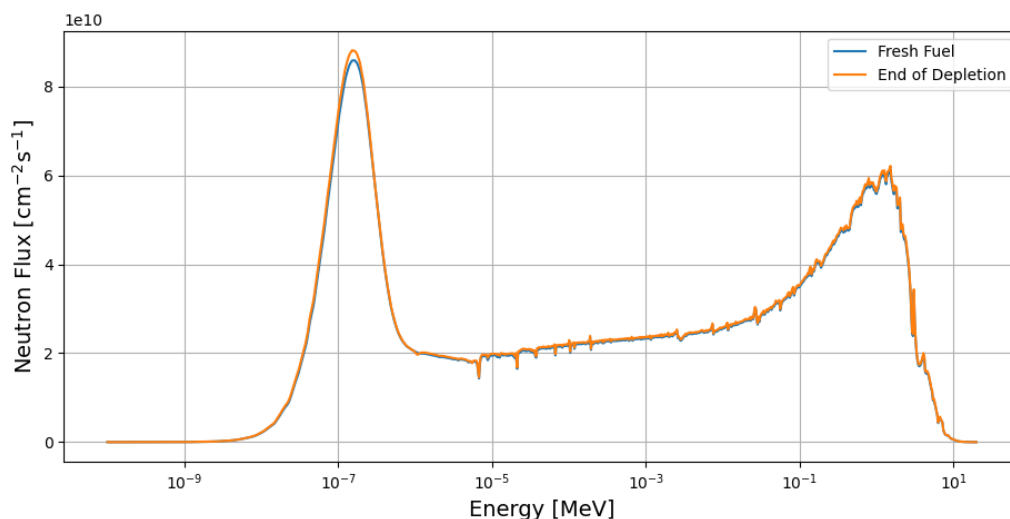


Figure 7-6: Snowflake neutron spectra with different fuel compositions

The difference in Snowflake neutron spectra (Figure 7-6) arising from depletion is similar to that of the GCR (Figure 6-8), although smaller in magnitude. Both models have similar average burnup, however, the GCR burnup has a greater maximum and smaller minimum. The greater number of fission products in regions of high flux most likely contribute to the larger differences in neutron spectrum over time in the GCR.

Another noticeable difference in neutron spectra evolution occurs around $2\text{E-}07$ MeV (0.2 eV). This microreactor has a lower fraction of thermal neutrons than the GCR, leading to smaller relative changes in the thermal neutron spectrum in Figure 7-7 relative to those in Figure 6-9.

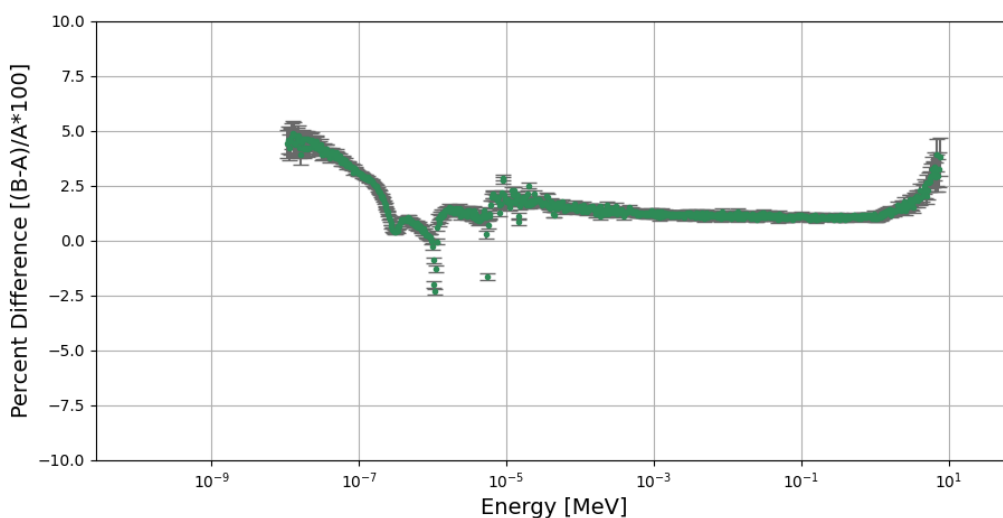


Figure 7-7: Snowflake neutron spectrum: depleted fuel percent difference from fresh fuel

Similar to the GCR, the neutron spectrum is very thermal, and error bars in Figure 7-7 result from propagating the statistical uncertainty through the percent difference calculation. The error bars represent the 95 percent confidence interval ($\pm 2\sigma$).

The different components of the Snowflake gamma spectrum are plotted in Figure 7-8. As with the GCR, the fresh fuel decay spectrum has a relatively low total intensity due to a lack of fission products. Additionally, differences in the fission spectrum (red and yellow lines) are small, and best seen in Figure A-2, which uses a linear y-axis.

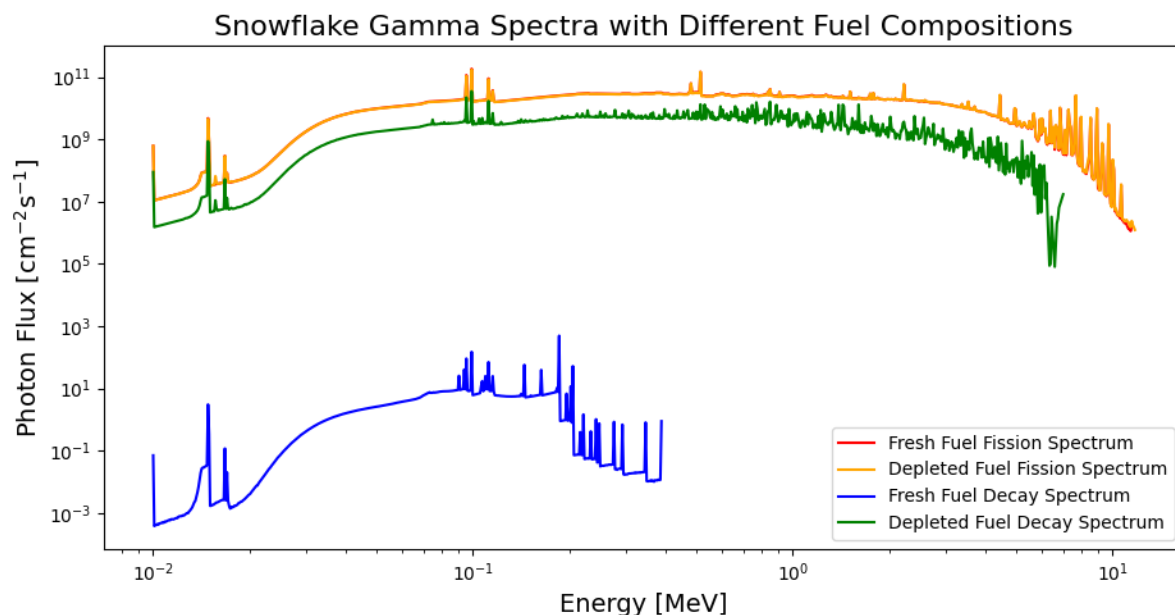


Figure 7-8: Snowflake gamma spectra with different fuel compositions (log y-axis). The two fission spectra are of the same magnitude and appear overlapped.

The gamma flux in the Snowflake microreactor is 1–2 orders of magnitude smaller than that of the GCR. This is roughly how much less uranium is in Snowflake. Because both reactors have TRISO fuel and a graphite monolith, the gamma peaks are very similar. Note that as discussed previously, the fresh fuel decay spectrum is several orders of magnitude lower than either the depleted fuel–decay spectrum or the fission spectra. A version of Figure 7-8 with a linear y-axis can be found in Appendix A.1.

7.5 Microreactor MC&A Considerations

The Snowflake microreactor discussed in this section is simulated with 19.5 percent enriched fuel, and at the BOL there is approximately 169 kg of uranium in the core. The information presented in this section is used to calculate the MC&A relevant parameters. Of note, as with the GCR, the fuel form for this design is fundamentally different than that of the MSR. Recall that the Snowflake microreactor design consists of TRISO compacts located in stacks that are within flakes (i.e., unit

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cells). As with the GCR, in addition to the total amount of material present, there are also different units which must be considered.

7.5.1 Microreactor SNM Quantities

The total amount of uranium at the BOL for this reactor design, as noted above, will be approximately 169 kg (with no initial plutonium present). At the EOL, the total amount of uranium will be approximately 163 kg and there will be approximately 0.9 kg of plutonium. Thus, there will be a change in uranium quantity of approximately 6 kg. However, as with a traditional facility, this difference is due to transmutation and is expected. It should be noted that the number of compacts will not change during the operating period, and there will be no exchange of compacts (i.e., operation will be static). Based on these considerations, the procedure for calculating the change in SNM quantities for this design is similar to what is discussed in Section 6.6.

Similar to the GCR, another MC&A-related parameter that is relevant is the amount of SNM per compact. At the BOL, each compact will contain 0.27 g of uranium and no plutonium. The EOL composition will depend on the compact location in the core because burnup will not be uniform, as discussed in Section 7.2. As an example, consider the compacts located in the highest burnup region. At the EOL, each compact will contain 0.26 g of uranium and 1.5E-3 g of plutonium. This results in 1 stack, with 68 compacts, containing approximately 18.0 g and 0.10 g of uranium and plutonium, respectively, at the EOL. Each of the unit cells contains 12 stacks, so for this region, the uranium and plutonium at the EOL for each unit cell will be approximately 216 g and 1.23 g, respectively. For these stacks to be treated as items, consideration would need to be given to the fact that an item should be a discrete amount of material that has a unique identification number.

7.5.2 Microreactor SNM Quantification Uncertainty

The uncertainty associated with the values calculated for the SNM quantities depends on what is tracked for the inventory, as discussed in Section 7.5.2 for the GCR design. However, the number of compacts present in this reactor is significantly fewer than in the GCR. Based on geometrical assumptions, the number of compacts in the Snowflake microreactor design is 24,480. For this system, a one percent counting error would lead to a discrepancy of approximately 65 g of uranium, using the EOL isotopics and assuming that all compacts were in the highest burnup region (used in the SNM calculation examples). If a one percent measurement were made on the mass, assuming all compacts were measured, the range of uranium values would span approximately 3 kg. Considerations related to different units for material accountancy are discussed in Section 7.5.3.

7.5.3 Microreactor Safeguards Discussion

The application of inventory difference calculations for this reactor design depends on what, if anything, is designated as an item. If this system is determined to be an item facility and the entire reactor (or the reactor core) is a unit, then the MC&A approach would likely rely predominately on verification of the item location. In this scenario, the expected difference in items between inventories would be zero. However, if a smaller unit was treated as an item, then the inventory

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requirements would have to be considered accordingly. For example, if each unit cell is an item, then a mechanism for individually identifying and counting the unit cells would be required. Again, in this scenario, the expected difference between inventories would be zero, but the inventory difference period would likely need to be evaluated. In the case of a microreactor with a long (multi-year) lifespan without refueling, performing a physical inventory of each item (unit cell, in this example) during the operating period could be unfeasible. However, it is helpful to note that at a currently operating nuclear power plant, the core is loaded and records maintained of the fuel location and identification. These records are relied on during physical inventory taking to declare the core inventory as the reactor is considered self-sealing while at power and/or while assembled.

Another potential approach is to identify operating parameters correlated to material inventory. This report does not explore the feasibility of this methodology.

8 Conclusions

Some advanced reactors present unique deployment considerations because of their non-traditional designs, fuel forms, and facility configurations. These considerations could include the identification and implementation of effective MC&A methods. The MC&A approach for traditional light water reactor designs is based on material accounting as an item facility, e.g., counting fuel assemblies. These approaches are well established for the current fleet of nuclear power plant facilities; however, they may not be applicable to all advanced reactor designs. This study considered three different advanced reactor designs: a liquid-fueled molten salt reactor, a gas-cooled reactor, and a microreactor. Models of the three designs were developed to demonstrate that simulations could be used to generate SNM information and the associated radiation signatures as a function of time.

Some of the advanced reactor designs, including those modeled for this project, may not lend themselves to traditional MC&A approaches. The liquid fuel of the MSR design may require MC&A approaches that differ from those with the fuel in a light water reactor. As with current reactors, under normal operating conditions, the SNM quantities will change, i.e., the total amount of uranium will decrease, the total amount of plutonium will increase, and the isotopics will change. These changes are expected, but the impact on potential signatures that could be leveraged for MC&A purposes must be understood. These considerations would benefit from additional research to fully understand the impacts on different MC&A approaches and their associated uncertainties. Another consideration for MSR designs is that a complete physical inventory of the SNM in the reactor is likely not feasible.

Thus, alternative approaches to traditional physical inventories could be considered. The implementation of such approaches will be enabled by further simulations and experimentation to identify the signatures that can be tracked and to fully understand their associated uncertainties.

TRISO-fueled systems also present a unique set of challenges related to MC&A, such as the microreactor and GCR concepts presented in this report. Although the fuel compacts could potentially be considered items and thus counted, the implementation of current MC&A methods requires some consideration. Note that an item is a discrete amount of material with a unique identification. For instance, at a fuel fabrication plant, a fuel pellet without an ID is generally not an item, but a container of fuel pellets with an identification number can be an item. The number of fuel compacts in TRISO-fueled systems is quite large (design dependent, but in some scenarios there could be several million in a core), which makes counting individual compacts challenging. Methods of automating the counting process require development and testing, particularly if a tracking technology is employed. Correlating an individual compact with its expected burnup may be important for SNM tracking purposes.

In some designs, compacts are dynamic during reactor operation. In others, including the two studied in this work, compacts are stationary. In this scenario, an additional consideration is the potential grouping of compacts into a larger item unit. If multiple compacts can be contained in a specified form, such as a stack, the number of individual items to be tracked for MC&A purposes could be reduced (analogous to tracking a fuel assembly versus individual pins). In this case, the different units that could be formed, and how inventory would be performed, needs to be established.

It should also be noted that a change in the isotopics will occur from normal reactor operations. If a tracking method or SNM quantification approach is desired to account for those changes dur-

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ing operation, this may require additional simulations and experimentation to establish how these changes correlate to operating parameters, the effects of uncertainties, and the impact on potential signatures.

Although there are numerous advanced reactor designs being explored with different challenges, similarities exist in the process required to develop an MC&A approach. For the three advanced reactor designs presented here, further model development and simulation efforts should consider the potential signatures that can be correlated to reactor operating parameters. A sensitivity study could also be performed to establish the material changes that are statistically identifiable based on different SNM signatures. The effect of model uncertainty will also require further assessment. This is particularly relevant when establishing use of models to generate predicted measurement parameters under various conditions. There are also uncertainties associated with different measurement techniques that will need to be considered for MC&A applications.

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Appendix A: Linear Y-Axis Gamma Spectrum Plots

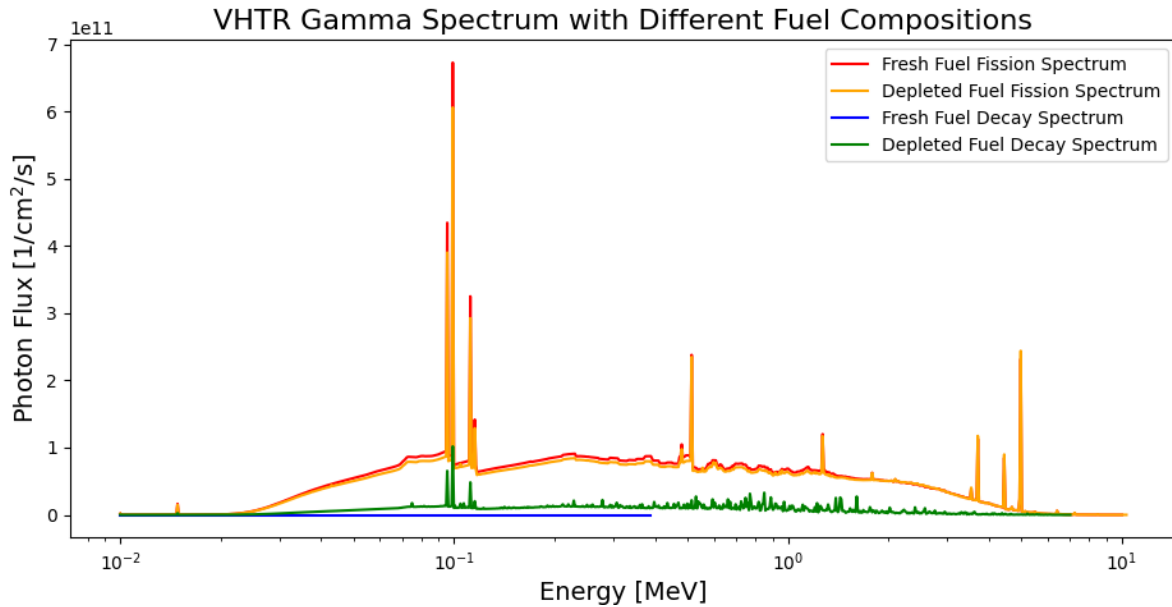


Figure A-1: VHTR gamma spectra with different fuel compositions (linear y-axis)

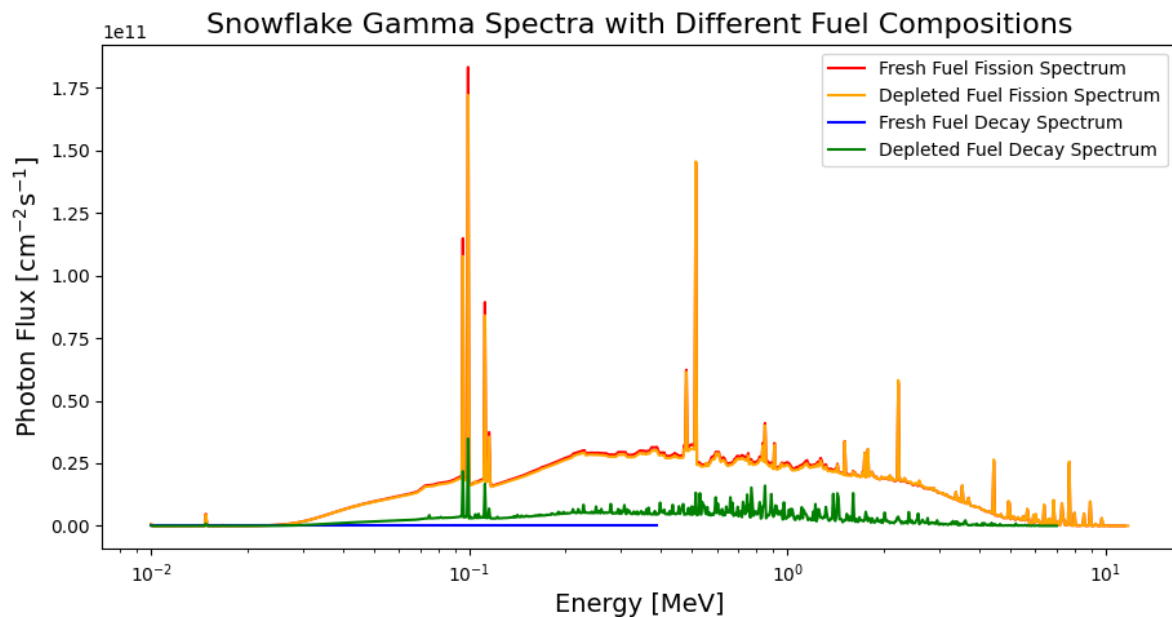


Figure A-2: Snowflake gamma spectra with different fuel compositions (linear y-axis)